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Chapter 1: INTRODUCTION

1.1 Background

Modern distributed computing environments have grown dramatically more complex. Today's systems often include hundreds of microservices and thousands of interconnected nodes deployed across multiple locations, each generating continuous streams of telemetry data that record internal and external behavior in real time. System logs represent one of the most important data streams. They provide the primary record of system operations, the first indication of performance degradation, and, when properly analyzed, a forensic trail to root causes before failures escalate into outages. He et al. (2020) found that fault-tolerant distributed systems can generate terabytes of logs daily. Operations teams face not a shortage of diagnostic information, but rather the challenge of processing and interpreting it as quickly as it arrives and at sufficient scale.

Automated log analysis has evolved from hand-crafted rules to statistical machine learning and, more recently, to deep learning and large language models. Early systems relied on manually specified regular expressions (Du et al., 2017). Machine learning brought data-driven detection. Xu et al. (2009) applied Principal Component Analysis to Hadoop Distributed File System logs. Liu et al. (2008) introduced Isolation Forest for unsupervised anomaly detection. These techniques improved coverage by learning from historical data rather than being limited to pre-defined patterns. Deep learning models like DeepLog (Du et al., 2017) and LogBERT (Guo et al., 2021) achieved F1-scores near 0.98 on standard datasets. Despite these advances, two problems persist: limited interpretability and inability to use external contextual knowledge at inference time.

1.2 Problem Statement

Existing automated log analysis systems struggle to achieve accurate, timely, and interpretable anomaly detection at the scale modern distributed infrastructure demands. To understand this problem, we must examine both its practical consequences and the specific technical limitations that sustain it.

The cost of failing to identify system anomalies in production environments is substantial. Gartner (2023) reports that unplanned IT infrastructure downtime costs

approximately \$5,600 per minute. Mean Time to Detect (MTTD)—the interval between fault occurrence and detection by monitoring systems—is a fundamental reliability metric (Patterson, 2002). In organizations using rule-based log analysis, MTTD can extend to tens of minutes or hours when failure modes fall outside pre-specified pattern libraries. Improved automated detection can reduce MTTD to seconds. In financial trading systems, healthcare platforms, or e-commerce environments, the difference between detection in seconds versus minutes can mean substantial data loss, regulatory exposure, and reputational damage.

Existing automated approaches fail to bridge this gap for three well-documented reasons:

1. Inflexibility and high false positive rates: Rule-based log analysis, the most widely deployed method in practice, matches log content against predefined pattern libraries. If a failure type was not anticipated when rules were created, it goes undetected, leading to false negatives and allowing failures to escalate. The high false positive rate, exceeding thirty percent in practice (Du et al., 2017), compounds this problem. Practitioners experience "alert fatigue"—a well-documented condition where genuine alerts are deprioritized or ignored (Sommer and Paxson, 2010). The maintenance burden aggravates these difficulties, as rule libraries must be manually updated as system architecture evolves.

2. Lack of interpretability: While machine learning and deep learning have improved accuracy in log data analysis, they introduce a new challenge: lack of interpretability. Approaches like Isolation Forest, PCA, and LSTM-based networks can classify log sequences as anomalous or normal and provide numerical scores, but offer no justification. For an operator receiving a warning based on a detected anomaly, an anomaly score provides little actionable information. There is no way to determine whether the anomaly relates to a network timeout, memory leak, or authentication problem. This directly impacts MTTD. Chen et al. (2023) argue that anomaly detection systems must be evaluated not only on accuracy but also on interpretability.

3. Knowledge isolation and hallucination: All approaches mentioned share the limitation of being closed systems—their analytical capabilities are limited to information

encoded in the system or programmed by rule authors. When interpreting a suspicious log pattern requires knowledge of a recently discovered vulnerability, a known interaction between system components, or a historical event in a similar system, these approaches cannot access that information. Large language models (LLMs) partially address this through broad knowledge encoded in model parameters, but they suffer from a well-known failure mode: hallucination. Ji et al. (2023) demonstrated that LLMs generate factually incorrect output with high confidence in technical domains.

The cumulative effect of these three limitations creates a monitoring environment where organizations face a three-part tradeoff: fast but often wrong, accurate but unclear, or seemingly competent but unreliable. No solution currently addresses all three issues adequately. Table 1.1 compares existing approaches with the proposed system on operational metrics relevant to system administrators.

Metric	Rule-Based	ML / Deep Learning	Proposed Agentic RAG
MTTD	Minutes–Hours	Seconds–Minutes	Seconds (target)
False Positive Rate	High (>30%)	Moderate (10–20%)	Low (target <10%)
Explainability	Partial	None	Full reasoning chain
Adapts to new patterns	No	Requires retraining	Yes (via retrieval)
External knowledge use	No	No	Yes (FAISS KB)
Alert actionability	Low	Low	High (with context)

Table 1.1: Operational comparison of existing log analysis approaches and the proposed system

What is required and has not been implemented or tested before—is a system that combines the contextual knowledge base of retrieval-augmented systems with the iterative and adaptive reasoning properties of agentic artificial intelligence. Such a system would draw upon external knowledge sources, adjust its analytical approach based on interim results, and produce outputs that are both correct and useful to those responsible for acting on them. This dissertation proposes and tests such a system.

1.3 Proposed Solution

This dissertation introduces an Agentic Retrieval-Augmented Generation (RAG) framework for system log analysis designed to overcome the three limitations identified in Section 1.2. The framework comprises four components. First, a log preprocessing module normalizes system log entries into structured formats using the Drain3 algorithm (He et al., 2017), parsing entries with severity levels, system components, and log templates while masking variable contents. Second, a vector-based knowledge base using the FAISS library stores domain knowledge documents, error patterns, and historical incident reports. Third, an agentic controller implements the ReAct pattern (Yao et al., 2023), capable of up to five iterative reasoning steps per log entry and autonomously deciding which process is needed at each step: retrieving knowledge, revising severity assessment, or producing a structured alert. Fourth, an alert generator produces human-readable structured alerts with complete reasoning chains, involved components, confidence levels, and recommended actions.

The proposed system differs from conventional RAG and other log analysis methodologies through its iterative reasoning capability. While standard RAG retrieves information and generates a response in a single step, the proposed system adapts its responses based on findings during the reasoning process. This mirrors the approach of an experienced system engineer who would iteratively reason about possible consequences and determine the most appropriate action. The system seeks to reduce MTTD through automated contextual analysis in seconds while addressing alert quality by providing context to operators, enabling them to take necessary action without further investigation.

1.4 Research Questions

The limitations identified in Section 1.2 and the solution proposed in Section 1.3 led to three research questions that frame the empirical investigation in this dissertation:

RQ1: Does the agentic RAG approach improve accuracy for anomaly detection in system log analysis compared to non-agentic RAG and rule-based approaches based on precision, recall, F1-score, and false positive rate on HDFS and BGL benchmark data?

RQ2: Which baseline system approach—rule-based, Isolation Forest, or non-agentic RAG—shows the greatest competitiveness compared to the proposed agentic RAG framework, and what aspects account for performance variations in detecting anomalies?

RQ3: Does the proposed agentic RAG system demonstrate greater alert specificity compared to the rule-based approach based on qualitative evaluation using a rubric measuring component identification, root cause explanation, and recommendation specificity?

Research Question 1 addresses effectiveness—whether multi-step agentic reasoning based on retrieved knowledge outperforms existing approaches for detection. Research Question 2 identifies which baseline approach best approximates the proposed system and why, helping determine whether the computational cost of agentic reasoning proves worthwhile or whether a simpler machine learning approach achieves similar results. Research Question 3 shifts focus to qualitative aspects of alert quality for operators affected by system-generated alerts, a factor not captured in quantitative precision and recall measures.

1.5 Aims and Objectives

The main aim of this dissertation is to design, implement, and evaluate an agentic RAG framework for system log analysis that overcomes the interpretability, flexibility, and knowledge grounding limitations of existing techniques, and to provide empirical evidence for the effectiveness of the proposed approach in comparison with three baselines.

Seven specific objectives follow from this aim:

O1: To conduct a critical review of the literature on system log analysis techniques, including rule-based systems, machine learning, deep learning, RAG, and agentic AI systems, and to identify the specific limitations of existing techniques that the proposed approach addresses.

O2: To develop an agentic RAG system architecture that addresses the three identified issues through a knowledge retrieval mechanism, a multi-step ReAct reasoning controller, and a structured alerts generator.

O3: To implement the proposed framework as a working research prototype using Python, LangChain, FAISS, Drain3, and Mistral 7B through Ollama.

O4: To implement three comparative baseline system prototypes: a rule-based keyword matcher system, an Isolation Forest classifier system, and a non-agentic single-pass RAG system.

O5: To evaluate all developed system prototypes on the HDFS and BGL datasets with standardized detection performance metrics (Precision, Recall, F1-score, False Positive Rate) and perform paired t-tests to determine statistical significance, addressing RQ1.

O6: To analyze the comparative detection performance results and determine which baseline system performs closest to the proposed framework, addressing RQ2.

O7: To analyze the quality of alerts generated by the agentic RAG system and the rule-based system by comparing a sample set using a standardized rubric to determine component identification, root cause explanation, and recommendation specificity, addressing RQ3.

1.6 Project Milestones

The dissertation follows the Waterfall Software Development Life Cycle, consisting of six milestones within eleven weeks (February to May 2026). This model ensures that every milestone builds upon the deliverables of the preceding one, which is necessary for a research project where evaluation accuracy depends on correct implementation. The six milestones are: literature review and system requirements (February), system architecture and baseline designs (early March), implementation of all systems (late March), validation and pilot experiments (early April), complete experiment results (late April), and final dissertation submission (May).

1.7 Dissertation Structure

The remainder of this dissertation is structured as follows. Chapter 2 reviews the literature on log analysis from rule-based and machine learning methods to deep learning, RAG, and agentic AI, concluding with the specific gap this research addresses. Chapter 3 discusses the research methodology, including the experimental design and mixed-method evaluation approach. Chapter 4 presents the system design and architecture. Chapter 5 discusses implementation and technologies utilized. Chapter 6 presents evaluation results for both quantitative detection metrics and qualitative alert evaluation. Chapter 7 discusses results in relation to all three research questions and the broader literature. Chapter 8 concludes the dissertation with contributions, limitations, and future work.

Chapter 2: LITERATURE REVIEW

2.1 Introduction

The literature on automated log analysis spans several decades and has evolved considerably in recent years, driven by advances in natural language processing and the availability of open benchmark datasets. Development has been particularly rapid since 2020, with transformer-based models, large language models, and retrieval-augmented generation entering the log analysis research agenda. This chapter reviews the relevant body of work across five areas: traditional log analysis methods, deep learning techniques, large language models, retrieval-augmented generation, and agentic AI. The chapter examines evaluation frameworks used to benchmark these approaches and concludes by identifying the specific gap this research addresses.

2.2 Rule-Based Log Analysis

Rule-based log analysis remains the most widely deployed category of automated monitoring in production environments, reflecting both its operational simplicity and the inertia of established tooling rather than technical superiority over recent approaches. The fundamental mechanism is straightforward: system administrators define patterns—typically regular expressions or keyword lists—that correspond to known failure signatures. When a log entry matches a pattern, an alert is raised. The approach requires no training data, executes in microseconds, and produces results that are, in principle, fully transparent and auditable.

He et al. (2020), in a comprehensive survey of automated log analysis techniques, characterize rule-based systems as occupying a necessary but fundamentally constrained role. Their central limitation is the closed-world assumption: the system can only detect what it has been explicitly instructed to look for. In fast-moving deployment environments where software components, infrastructure configurations, and operational patterns evolve continuously, novel failure modes—precisely those most likely to be consequential—pass entirely undetected. Maintaining comprehensive and current rule libraries becomes a substantial operational burden, and available evidence suggests most organizations fall short of complete coverage.

Le and Zhang (2022) provide direct empirical evidence of this limitation through systematic evaluation of log parsing and analysis methods across multiple datasets exhibiting label noise, format drift, and template evolution—conditions representative of live production systems. Their findings demonstrate that rule-based approaches experience consistent performance degradation as log formats evolve, and that the gap between benchmark results on clean datasets and real-world performance is considerably wider for rule-based systems than for data-driven alternatives.

The false positive problem compounds these concerns. Keyword-based pattern matching cannot reliably distinguish between a log entry containing ERROR as part of a genuine unhandled fault and one containing it as part of a successfully handled exception. When false positive rates are high—empirical studies suggest rates exceeding thirty percent in production environments (He et al., 2020)—the practical consequence is alert fatigue: progressive deterioration of operator responsiveness to genuine alerts. Alert fatigue is not merely an inconvenience; it is a safety-critical failure mode that can cause genuine high-severity incidents to be missed or deprioritized precisely because they arrive in a stream of predominantly spurious notifications.

2.3 Machine Learning Approaches

The limitations of rule-based systems motivated development of data-driven log analysis methods capable of learning to distinguish normal from anomalous behavior without requiring explicit pattern specification. The machine learning generation introduced clustering, dimensionality reduction, and ensemble anomaly detection, all addressing the rigidity problem of rule-based approaches by adapting their behavior to statistical properties of the log data they were trained on. However, this generation introduced its own distinct set of limitations.

He et al. (2020) survey this generation extensively, noting that methods based on statistical features—such as log event frequency counts, inter-arrival time distributions, and template co-occurrence patterns—can identify statistical outliers that escape keyword-based detection. The principal advantage is adaptability: by learning a representation of normal system behavior from historical log data, these methods can detect deviations even when the specific failure signature has not been observed before.

This makes them structurally better suited to novel anomaly detection than rule-based systems, at least in principle.

In practice, however, machine learning approaches introduce two limitations that have constrained their adoption. The first is the feature engineering bottleneck. These methods require conversion of raw, unstructured log text into numerical feature vectors, and detection quality is directly dependent on feature quality. He et al. (2020) note that naively constructed features—such as simple term frequency representations—fail to capture semantic relationships and sequential dependencies that characterize log data, resulting in detection models sensitive to superficial statistical patterns rather than underlying operational semantics. The second limitation is more fundamental: machine learning methods produce anomaly scores or binary classification outputs with no accompanying explanation. An anomaly score of 0.87 provides an operator with no information about which component generated the anomalous behavior, what the likely cause might be, or what remediation steps would be appropriate. Le and Zhang (2022) identify this interpretability gap as one of the most persistent unresolved challenges in the log analysis field.

A further concern specific to unsupervised approaches—which represent the majority of deployed machine learning log analysis systems—is calibration sensitivity. If training data contains anomalies mislabeled as normal, which is common in environments where historical incidents were not systematically documented, the learned model of normal behavior will be corrupted. Chen et al. (2023) discuss this as a calibration problem that complicates the reliability of machine learning anomaly scores in operational contexts.

2.4 Deep Learning Methods for Log Analysis

Deep learning approaches emerged as a response to the feature engineering bottleneck of classical machine learning methods. Rather than relying on hand-crafted numerical representations, deep learning models learn representations directly from raw log sequences, capturing both semantic content of individual log events and sequential dependencies between them. This generation achieved substantially higher detection accuracy on standard benchmarks and reduced manual effort required for feature design,

but introduced its own limitations around robustness to real-world conditions and the persistent absence of interpretable outputs.

Guo et al. (2021) introduced LogBERT, applying the BERT transformer architecture to log anomaly detection through a two-stage approach: unsupervised pre-training on unlabeled log sequences, followed by anomaly detection using the trained representations. LogBERT demonstrated F1-scores of 0.982 on the HDFS dataset and 0.908 on the BGL dataset, representing meaningful improvement over earlier methods and establishing transformer-based representations as the state of the art for log anomaly detection on standard benchmarks. The innovation was using self-supervised pre-training to develop semantic representations of log templates that capture contextual relationships between log events—relationships that feature-based methods cannot represent.

Zhang et al. (2024) contribute complementary work with LILAC, a log parsing approach that uses large language models with an adaptive parsing cache to improve the quality of structured log representations that downstream analysis methods depend on. Their work demonstrates that transformer-based models can contribute to the log preprocessing stage as well as the analysis stage, and that log template extraction quality—a step often treated as solved in the log analysis literature—has substantial downstream impact on anomaly detection performance. This finding is directly relevant to the present research, which uses the Drain3 parsing algorithm for preprocessing.

Despite these accuracy improvements, Le and Zhang (2022) provide a sobering assessment of deep learning methods under realistic operating conditions. In systematic evaluation across multiple datasets with varying degrees of label noise, format drift, and template evolution, they find that performance degrades significantly relative to clean-benchmark results. Methods achieving near-perfect F1-scores on standard HDFS evaluations demonstrate substantially reduced performance when evaluated on logs exhibiting the kinds of variation common in live environments. This finding is important for interpreting published benchmark results: high accuracy figures reported in many papers may reflect performance on idealized evaluation conditions rather than genuine operational capability.

Most critically, the interpretability limitation remains entirely unresolved across all deep learning approaches. Transformer models can be probed using attention visualization and gradient-based attribution methods, but these techniques do not produce the kind of natural language explanation that operations engineers require to make rapid, confident triage decisions. Chen et al. (2023) argue that the gap between detection accuracy and decision support quality represents one of the most significant barriers to practical deployment of deep learning methods in operational monitoring, and that future log analysis research must treat interpretability as a primary design requirement rather than an optional supplement to detection performance.

2.5 Large Language Models in Log Analysis

The emergence of large language models as general-purpose reasoning systems opened a new line of inquiry for log analysis researchers. Unlike their predecessors, LLMs can generate natural language explanations of their outputs, reason over unfamiliar input patterns, and use broad technical knowledge acquired during pre-training on large corpora of technical documentation, academic papers, and software engineering content. These properties directly address the interpretability limitations of machine learning and deep learning methods, making LLMs attractive candidates for log analysis applications where human-readable output is required. However, LLMs introduce a qualitatively different class of failure mode that demands careful consideration before deployment in reliability-critical contexts.

Ji et al. (2023), in a comprehensive survey of hallucination in natural language generation systems, document that LLMs produce factually incorrect outputs with high confidence across a wide range of domains, with particularly pronounced failure rates in technical and scientific contexts. For log analysis applications, the implications are substantial. A system that confidently misdiagnoses a production fault—attributing a network partition to database connection pool exhaustion, for example—may actively impede incident resolution by directing engineering attention toward the wrong component and delaying correct diagnosis. Ji et al. (2023) argue that hallucination in LLMs is not a surface-level deficiency that can be eliminated through prompt engineering, but a structural

consequence of how parametric knowledge is encoded in model weights—knowledge that is inherently imprecise, potentially outdated, and unverifiable at inference time.

Zhang et al. (2024) demonstrate that LLMs can be adapted effectively for the upstream log parsing task when combined with an adaptive caching mechanism that stores and reuses previously parsed templates. Their LILAC framework achieves parsing accuracy competitive with dedicated log parsing tools while providing greater flexibility in handling previously unseen log formats. This finding is relevant because it illustrates that LLMs can contribute at multiple stages of the log analysis pipeline, not only at the analysis and alert generation stage.

Qi et al. (2023), in an empirical study of LLM-based log anomaly detection using ChatGPT-style models, find that prompt engineering and in-context learning techniques can improve accuracy of LLM outputs on log classification tasks, but that performance remains sensitive to prompt design and deteriorates on log formats that differ substantially from those encountered during pre-training. These findings collectively suggest that LLMs are a promising but structurally insufficient basis for standalone log analysis systems: their interpretability and reasoning capabilities are genuinely valuable, but their hallucination risk and sensitivity to input distribution shift are genuine and non-trivial limitations.

2.6 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) was introduced by Lewis et al. (2020) as a framework for combining the generative capabilities of large language models with the precision of information retrieval from external knowledge sources. The foundational insight is that LLM generation need not rely solely on parametric knowledge encoded in model weights—knowledge that may be outdated, imprecise, or absent for highly specialized domains. By retrieving relevant documents from an external knowledge base at inference time and conditioning generation on those documents, RAG grounds the model's outputs in verifiable evidence, substantially reducing hallucination rates on knowledge-intensive tasks.

Lewis et al. (2020) demonstrate across multiple question-answering benchmarks that RAG consistently outperforms both standalone retrieval systems and standalone LLMs, and that the combination produces outputs more factually accurate than either component in isolation. The retrieval component—implemented using dense passage retrieval with FAISS vector indexing—enables the system to locate relevant documents even when the query and target document do not share exact keyword overlap, a capability particularly valuable for log analysis where natural language description of a failure pattern may differ substantially from technical terminology used in the log entry itself. The generation component, conditioned on retrieved passages, produces outputs that reflect specific operational context rather than generic parametric knowledge.

Gao et al. (2023) provide a comprehensive taxonomy of RAG architectures that distinguishes between three levels of architectural sophistication. Basic RAG employs a single retrieval step followed by a single generation pass—a design that is simple and computationally efficient but may be insufficient for queries where relevant knowledge cannot be identified from the initial query alone. Advanced RAG introduces pre-retrieval query reformulation and post-retrieval re-ranking, improving retrieval quality at the cost of additional processing. Modular RAG supports iterative and adaptive retrieval strategies in which the system can retrieve additional information at multiple points during generation, guided by the content of previously retrieved passages. Gao et al. (2023) demonstrate that this architectural progression produces consistent improvements on tasks requiring multi-hop reasoning, which directly motivates the agentic multi-step design adopted in the present research.

Asai et al. (2023) take this reasoning further with Self-RAG, a framework in which a language model learns to adaptively control its own retrieval behavior—deciding, at each point during generation, whether additional retrieval is warranted based on what has been generated so far. Their experiments demonstrate that adaptive retrieval improves both accuracy and efficiency of RAG systems on tasks with heterogeneous knowledge requirements. The agentic controller in the present research implements a conceptually similar adaptive logic through the ReAct reasoning loop: at each step, the controller assesses whether additional retrieval is warranted before proceeding.

Es et al. (2023) introduce RAGAS, a framework for automated evaluation of RAG systems that assesses three dimensions of output quality: faithfulness, which measures the degree to which generated outputs are supported by retrieved documents; answer relevance, which measures the relevance of the generated response to the original query; and context relevance, which measures the relevance of retrieved documents to the query. RAGAS provides a principled basis for evaluating not just detection accuracy of the proposed RAG-based log analysis system, but the quality of the reasoning chain that connects retrieved knowledge to generated outputs—directly informing the evaluation methodology adopted for RQ3 of the present research.

2.7 Agentic AI and Multi-Step Reasoning

Agentic AI systems represent a qualitative extension of standard LLM capabilities that has attracted substantial research attention since 2022. Where a standard LLM generates a response to a prompt in a single forward pass, an agentic system executes an iterative loop in which the model reasons about what action to take, executes that action, observes the result, updates its reasoning state, and decides whether to continue or conclude. This architecture enables adaptive, multi-step problem solving under uncertainty that is not accessible to single-pass systems regardless of their scale or the sophistication of their prompting.

Yao et al. (2023) introduced ReAct, Reasoning and Acting, as a framework for interleaving chain-of-thought reasoning with concrete action execution in language models. The central contribution of ReAct is the demonstration that reasoning and acting are mutually reinforcing: chain-of-thought reasoning improves the quality of action selection by making the model's analytical state explicit before each action, while observations from executed actions provide new information that refines subsequent reasoning steps. In a series of multi-hop question-answering and interactive decision-making benchmarks, ReAct outperformed both standard prompting and chain-of-thought prompting without action capabilities, with improvements of up to 34% on tasks requiring multi-step information retrieval. The implications for log analysis are direct: a log sequence that cannot be classified from a single retrieval step because relevant

context is distributed across multiple knowledge sources is exactly the kind of task that ReAct's interleaved reasoning-and-acting architecture is designed to handle.

Schick et al. (2023) introduced Toolformer, demonstrating that language models can learn to use external tools—including search APIs, calculators, and structured databases—in a self-supervised manner, without requiring explicit tool-use supervision. By including tool calls within training data and training the model to predict when and how to invoke tools based on context, Toolformer models acquire the ability to retrieve and incorporate external information at the points where it is most needed. While the present research does not use Toolformer directly, its findings establish the feasibility of autonomous tool use in LLM-based systems and provide theoretical grounding for the design principle of the agentic controller—that knowledge retrieval should be driven by the model's own assessment of what information is needed at each reasoning step.

Wang et al. (2023), in a comprehensive survey of LLM-based autonomous agents, identify memory, planning, and action as the three core capability dimensions that determine agent performance. Memory encompasses both short-term context—the current reasoning state—and long-term storage through external knowledge bases. Planning encompasses task decomposition and self-reflection—the capacity to critique and revise earlier reasoning steps in light of new evidence. Action encompasses the set of tools and external operations available to the agent. The agentic controller in the present research implements each of these dimensions: short-term memory through the reasoning chain data structure, long-term memory through the FAISS knowledge base, planning through the ReAct loop with up to five reasoning steps, and action through five defined action types. Wang et al. (2023) document that this combination of capabilities enables agentic systems to handle tasks of significantly greater complexity than single-pass systems, particularly in domains where relevant information is distributed and the correct analytical path is not apparent from the initial query alone.

It is important to acknowledge that agentic AI introduces practical challenges that must be managed carefully. Longer inference chains increase computational cost, as each reasoning step requires an independent LLM call. Error propagation is a documented concern: incorrect reasoning in early steps can lead subsequent steps to build on faulty

premises, compounding rather than correcting the initial mistake. Wang et al. (2023) also note that evaluating agentic systems is considerably more complex than evaluating single-pass models, because performance depends not only on the quality of the final output but on the quality of the intermediate reasoning trajectory.

2.8 Evaluation Frameworks

The evaluation of log analysis systems requires careful attention to both dataset selection and metric design, since choices in both dimensions can substantially influence conclusions drawn. He et al. (2020) introduced Loghub, a collection of nineteen system log datasets from diverse computing environments including distributed systems, supercomputers, and operating systems. Two datasets from Loghub have become the dominant benchmarks in the log analysis literature and are adopted for evaluation in the present research. The HDFS dataset contains approximately eleven million log entries from a Hadoop Distributed File System cluster, with ground-truth anomaly labels at the block level. The BGL dataset contains approximately 4.7 million entries from a BlueGene/L supercomputer, with message-level labels indicating hardware and software faults of various types. Together, these datasets provide sufficient scale and label coverage for statistically meaningful comparative evaluation, and their widespread use in the literature makes results directly comparable to prior published work.

Le and Zhang (2022) provide an important methodological caveat regarding benchmark evaluation that is directly relevant to the design of the present research. In systematic evaluation of log parsing methods across multiple datasets, they demonstrate that reported performance figures are sensitive to sampling strategy, the proportion of anomalous examples in the test set, and the specific train/test split adopted. Methods that appear to perform similarly under idealized evaluation conditions can diverge substantially when evaluated under more demanding conditions—a finding that motivates the use of stratified sampling and repeated experimental trials in the present research, alongside paired t-tests for statistical significance testing.

Standard detection metrics—precision, recall, F1-score, and false positive rate—provide the quantitative backbone of performance comparison across the four systems evaluated in this research. However, Chen et al. (2023) make a compelling case that these metrics

are insufficient for evaluating systems intended to support human decision-making in operational monitoring contexts. A system that achieves a high F1-score but generates alerts that provide no actionable guidance is of limited practical value, regardless of its technical detection accuracy. Chen et al. (2023) propose supplementary evaluation protocols that assess the completeness and actionability of generated explanations, directly informing the qualitative evaluation rubric adopted for RQ3.

Es et al. (2023), through the RAGAS framework, contribute an automated evaluation methodology for RAG-specific quality dimensions that extends beyond detection metrics. RAGAS assesses faithfulness—the degree to which generated outputs are supported by retrieved documents rather than derived from unconstrained parametric knowledge—alongside answer relevance and retrieval precision. Faithfulness is particularly pertinent to the present research because it provides a proxy for the hallucination risk identified as a critical limitation of standalone LLM approaches. A high faithfulness score provides evidence that the proposed system's outputs are grounded in retrieved operational knowledge rather than fabricated.

2.9 Summary and Identified Research Gap

The review conducted in this chapter reveals a consistent and progressive trajectory in log analysis research: each successive generation of methods has improved upon its predecessors in specific dimensions while introducing new limitations, and no single approach has yet addressed the complete set of requirements that effective operational log monitoring demands. Table 2.1 summarizes this trajectory and maps each identified limitation to the component of the present research designed to address it.

Approach	Key Strength	Key Limitation	Gap Addressed by This Research
Rule-Based	Fast, transparent	Rigid, high FPR, no context	Knowledge retrieval + adaptive reasoning
Machine Learning	Learns from data	No interpretability, feature-dependent	LLM-generated reasoning chains
Deep Learning	High benchmark accuracy	Degrades in real conditions; opaque	Grounded generation via RAG
LLMs (standalone)	Interpretable outputs	Hallucination; no	Retrieval grounding

		domain grounding	reduces hallucination
Standard RAG	Grounded generation	Single-step; insufficient for complex logs	Multi-step agentic reasoning loop

Table 2.1: Summary of log analysis approaches, limitations, and research response

Rule-based systems remain operationally prevalent but are structurally incapable of adapting to novel failure modes, generate high false positive rates, and require continuous manual maintenance. Machine learning methods address the rigidity problem but introduce an interpretability gap that prevents their outputs from guiding effective operator action. Deep learning methods achieve high accuracy under idealized benchmark conditions but degrade under realistic conditions and deepen the interpretability problem rather than resolving it. Large language models contribute interpretable natural language outputs and broad technical knowledge but introduce hallucination risk that is structurally problematic in reliability-critical monitoring contexts. Standard RAG reduces hallucination through retrieval grounding but is architecturally constrained to single-step reasoning, which may be insufficient for the multi-faceted interpretation of complex log sequences.

Agentic AI—and specifically the ReAct architecture of Yao et al. (2023)—provides the theoretical and empirical basis for a system that can overcome the limitations of single-step RAG through iterative, adaptive reasoning. Wang et al. (2023) document substantial performance improvements from agentic designs over single-pass alternatives across a range of complex reasoning tasks. However, as both Wang et al. (2023) and Yao et al. (2023) acknowledge, agentic AI systems have to date been evaluated primarily in web navigation, question answering, and code generation tasks. There is no published research that has designed, implemented, and empirically evaluated an agentic RAG framework specifically for system log analysis and anomaly detection—comparing it systematically against rule-based, classical machine learning, and standard RAG baselines on established benchmark datasets with ground-truth labels.

This gap is precisely what the present research addresses. The following chapter describes the methodology through which that gap is filled: the comparative experimental

design, the dataset selection rationale, the evaluation metrics, and the development lifecycle adopted for implementing and testing all four systems.

Chapter 3

METHODOLOGY

3.1 Introduction

This chapter describes the methodological decisions that govern how the research questions presented in Chapter 1 are investigated. It begins by articulating the philosophical position that underpins the inquiry before justifying the choice of comparative experimental design as the overarching research strategy. The chapter then specifies the variables under investigation, describes the software development lifecycle adopted for building the systems under comparison, and provides a detailed account of the dataset selection rationale, evaluation metrics, and statistical approach used to interpret results.

3.2 Research Philosophy and Paradigm

The research is grounded in a positivist philosophical tradition, which holds that knowledge is best produced through objective, empirically verifiable evidence obtained under controlled conditions (Creswell and Creswell, 2022). This position is appropriate for the present study because the research questions are fundamentally quantitative: they ask whether one computational approach produces measurably better outcomes than others, and they can be answered through numerical metrics derived from controlled experiments. The ontological stance is one of objective realism—the phenomena under investigation, namely anomaly detection accuracy and alert quality, are assumed to exist independently of the researcher and to be amenable to measurement. The epistemological stance is empiricist: claims are validated through observation rather than theory alone.

A purely positivist approach does require qualification in the context of RQ3, which asks about the quality of alert outputs—a dimension that cannot be reduced to a single numerical metric without interpretive judgement. To accommodate this, the methodology incorporates a structured qualitative assessment of alert content alongside the quantitative detection experiments. This mixed-method approach—predominantly quantitative, with a qualitative component for alert quality evaluation—is consistent with Saunders et al. (2023), who argue that pragmatic mixed-method designs are appropriate when a single paradigm cannot fully address all dimensions of a research question.

3.3 Research Design

The primary research design adopted is a comparative experimental framework, in which multiple systems—the proposed agentic RAG framework and three baseline systems—are evaluated on the same inputs under identical conditions, and their outputs are compared using pre-specified metrics. This design is well-suited to RQ1 and RQ2, both of which require direct, evidence-based comparison of system performance across approaches. Comparative experimental design enables the isolation of the independent variable—the choice of log analysis approach—while controlling for confounding factors such as dataset characteristics, evaluation procedure, and implementation quality. Where confounding cannot be fully eliminated, the use of statistical significance testing provides a mechanism for distinguishing genuine performance differences from noise in the experimental results (Field, 2024).

This design has strong precedent in the log analysis literature. He et al. (2020) employ comparative experimental evaluation across multiple approaches and datasets as the standard methodology for assessing log analysis systems, and Le and Zhang (2022) extend this tradition by emphasizing the importance of evaluation under realistic conditions with stratified sampling and multiple experimental runs. The present research follows both precedents: all four systems are evaluated on the same HDFS and BGL test sets, produced by the same stratified 70/30 split, with statistical significance testing applied to validate observed differences. For RQ3, which requires qualitative assessment of alert content, the comparative framework is extended through the structured rubric

evaluation, maintaining the comparative logic while accommodating the inherently interpretive nature of alert quality assessment.

3.4 Research Variables

The research variables are defined with sufficient precision to enable unambiguous operationalization in the experimental design. Table 3.1 summarizes all variable types, their specific instantiation in this research, and the manner in which each is operationalized.

Variable Type	Variable	Operationalization
Independent	Log analysis approach	Four systems: Rule-Based, Isolation Forest, Non-Agentic RAG, Agentic RAG
Dependent (Detection)	Precision, Recall, F1-score, FPR	Computed from predicted vs. ground-truth labels on HDFS and BGL test sets
Dependent (Quality)	Alert actionability score	Rubric score (0-15) across three dimensions: component ID, root cause, recommendation
Control	Dataset, train/test split, implementation	Same datasets, identical 70/30 stratified split, documented hyperparameters

Table 3.1: Research variables and their operationalization

The independent variable—the log analysis approach—takes four values corresponding to the four systems under evaluation. Each system receives identical input in the form of preprocessed log entries from the test set and produces a binary classification output—anomalous or normal—alongside, for the two RAG-based systems, a textual explanation of the classification decision.

The dependent variables are divided into two categories. Detection performance variables—precision, recall, F1-score, and false positive rate—are computed from the binary classification outputs against ground-truth labels. Alert quality, the dependent variable for RQ3, is operationalized through the structured rubric score. Control variables ensure that observed differences in dependent variables can be attributed to the

independent variable rather than extraneous factors. Using identical datasets, the same train/test split, and documented hyperparameters for all systems introduces a deliberate conservatism that prioritizes internal validity.

3.5 Development Methodology

The software systems under comparison are developed and managed using the Waterfall Software Development Lifecycle. This choice merits explicit justification, since Agile methodologies have become the dominant framework for software development.

However, Agile's suitability depends on specific contextual factors that do not apply uniformly to all software development endeavors, and careful analysis of the present project's characteristics leads to the conclusion that Waterfall is the more appropriate choice.

Agile methodologies are designed to accommodate evolving requirements through iterative delivery and frequent stakeholder feedback. Their value is greatest in environments where requirements are ambiguous, users are actively involved in shaping the product incrementally, and deployment to end-users is the primary goal. The present project does not exhibit these characteristics. The research questions, datasets, evaluation metrics, and system components are all fully specified in advance. There are no end-users providing iterative feedback on functional requirements. The deliverable is not a deployed product but a research prototype evaluated against pre-specified performance criteria. In this context, the iterative flexibility that makes Agile valuable in commercial product development provides no particular benefit.

The Waterfall model organizes development into clearly bounded sequential phases—requirements, design, implementation, testing, evaluation, and documentation—each producing a specific deliverable and providing a verified foundation for the next. Table 3.2 summarizes the six Waterfall phases and their associated timelines and deliverables.

Phase	Timeline	Key Deliverable
Requirements	Feb 16–25	Literature review, system requirements, preprocessed datasets, configured environment
Design	Feb 26–Mar 7	System architecture, component

		interfaces, knowledge base schema, baseline designs
Implementation	Mar 8–31	Functional agentic RAG prototype, all three baseline systems, integration test results
Testing	Apr 1–10	Unit and integration test reports, validated preprocessing pipeline, pilot experiment results
Evaluation	Apr 11–20	Comparative performance results, statistical analysis report, visualizations
Documentation	Apr 21–30	Complete dissertation manuscript, formatted bibliography, submission-ready PDF

Table 3.2: Waterfall development phases, timelines, and deliverables

3.6 Dataset Selection and Preparation

Two publicly available, labeled log datasets are selected for evaluation: the Hadoop Distributed File System (HDFS) dataset and the BlueGene/L (BGL) supercomputer dataset. Both are drawn from the Loghub benchmark collection maintained by He et al. (2020), which provides standardized versions of both datasets alongside detailed descriptions of their provenance and label generation procedures. The selection of these two datasets is justified on four grounds.

First, both datasets provide ground-truth anomaly labels, which are a prerequisite for quantitative evaluation of detection performance. The absence of ground-truth labels is one of the most common barriers to rigorous evaluation in the log analysis literature. Second, both datasets are sufficiently large to support statistically meaningful evaluation: the HDFS dataset contains approximately eleven million log entries with block-level anomaly labels, while BGL contains approximately 4.7 million entries with message-level fault labels. Third, both datasets are widely used as benchmarks in the log analysis literature, which means that results obtained here can be directly compared to published figures. Fourth, the two datasets differ significantly in their structural characteristics,

which provides the basis for the cross-dataset comparison motivating RQ2: HDFS logs are generated by a distributed storage system and exhibit block-level failure patterns driven by data replication faults, whereas BGL logs originate from a supercomputer and exhibit hardware and system fault patterns with distinct temporal signatures. This structural heterogeneity means that findings consistent across both datasets carry greater external validity than findings derived from a single dataset alone.

Dataset preparation follows a standardized procedure for all four systems to ensure comparability. Raw log files are parsed using the Drain3 algorithm with a fixed-depth tree of depth four and a similarity threshold of 0.4. A 70/30 stratified train/test split is applied, where stratification is performed on the anomaly label to ensure that the proportion of anomalous examples in the test set reflects their proportion in the full dataset. This is particularly important for the HDFS and BGL datasets, in which anomalous logs represent a minority class.

3.7 Evaluation Strategy and Metrics

The evaluation strategy combines quantitative detection performance measurement, statistical significance testing, and structured qualitative assessment, with each component addressing one or more of the three research questions. Table 3.3 summarizes the evaluation metrics, the research question each addresses, and the rationale for its inclusion.

Metric	Addresses	Rationale
Precision	RQ1	Proportion of flagged anomalies that are genuine; low precision indicates high false positive rate
Recall	RQ1	Proportion of genuine anomalies detected; low recall indicates missed faults
F1-score	RQ1	Harmonic mean of precision and recall; primary summary metric for imbalanced datasets
False Positive Rate	RQ1	Proportion of normal logs incorrectly flagged; directly

		operationalizes alert fatigue risk
Alert Actionability	RQ3	Qualitative rubric score assessing component identification, root cause, and recommendation
Paired t-test p-value	RQ1, RQ2	Statistical significance of differences between systems; threshold $p < 0.05$

Table 3.3: Evaluation metrics, research question mapping, and rationale

3.7.1 Quantitative Detection Metrics

Precision, recall, F1-score, and false positive rate are computed from the standard confusion matrix derived by comparing each system's binary classification outputs against the ground-truth labels on the test set. F1-score is designated as the primary summary metric because it provides a single measure that reflects both precision and recall simultaneously—a property particularly important for imbalanced datasets where accuracy alone can be misleading. False positive rate is included as a secondary metric because it directly operationalizes the alert fatigue risk identified in Chapter 1 as one of the primary practical limitations of rule-based systems.

3.7.2 Statistical Significance Testing

Paired t-tests are used to determine whether observed performance differences between systems are statistically significant or attributable to random variation in the evaluation sample. The test is applied pairwise—comparing the proposed agentic RAG framework against each of the three baselines on each dataset—using per-sequence F1-scores as the unit of analysis. A significance threshold of $p < 0.05$ is adopted, consistent with standard practice in the systems evaluation literature (Field, 2024). For multiple pairwise comparisons within each dataset, the Bonferroni correction is applied to control the family-wise error rate.

3.7.3 Qualitative Alert Quality Evaluation (RQ3)

The qualitative evaluation of alert quality—required to address RQ3—is conducted using the structured rubric presented in Table 3.4. The rubric was developed by adapting the explainability evaluation criteria proposed by Chen et al. (2023), who argue that anomaly

detection systems must be assessed on the quality of their explanatory outputs as well as their detection accuracy. Three dimensions are assessed: component identification, root cause explanation, and recommendation specificity.

Dimension	Max Score	Scoring Criteria
Component Identification	5	5 = specific component named and contextualized; 3 = component named; 1 = generic; 0 = absent
Root Cause Explanation	5	5 = causal chain explained with evidence; 3 = likely cause stated; 1 = fault category; 0 = absent
Recommendation Specificity	5	5 = concrete actionable steps; 3 = general guidance; 1 = boilerplate; 0 = absent
Total	15	Score of 12+ indicates high actionability; 7-11 moderate; below 7 low

Table 3.4: Alert quality evaluation rubric for RQ3

The evaluation sample consists of twenty alerts drawn from each of the two systems compared under RQ3—the proposed agentic RAG framework and the rule-based baseline—for a total of forty alerts. The twenty alerts per system are selected by stratified random sampling from the test set anomaly outputs, stratified by severity level. Each alert is scored independently against the three rubric dimensions.

3.8 Technology Stack Justification

All systems are implemented in Python 3.12, which provides the most extensive ecosystem of machine learning, natural language processing, and data analysis libraries and is the de facto standard for research prototyping in these domains. The specific library choices are justified with reference to the functional requirements of the system and the constraints of a single-developer academic project.

Log parsing is performed using Drain3—the Python implementation of the Drain log parsing algorithm. Drain3 is selected because it operates online, processing logs

incrementally rather than requiring full dataset ingestion before parsing begins. The vector knowledge base is implemented with FAISS (Facebook AI Similarity Search), specifically the IndexFlatL2 index type, using sentence embeddings generated by the all-MiniLM-L6-v2 model from the sentence-transformers library. FAISS is chosen for its computational efficiency—exact nearest-neighbor search completes within the 500ms latency budget—and for its broad adoption in RAG system implementations.

The LLM component uses Mistral 7B Instruct, served locally via Ollama. The choice of a locally-served open-source model rather than a commercial API such as OpenAI GPT-4 is motivated by three considerations. First, local inference ensures complete reproducibility: the model version and inference parameters are fixed and under the researcher's direct control. Second, using a 7B parameter model constrains the performance of the LLM component in a way that is consistent with the research question: the goal is to demonstrate that the agentic reasoning architecture adds value, not merely that a larger model produces better outputs. Third, local inference eliminates per-token API costs.

3.9 Ethical Considerations

The research raises three categories of ethical consideration. First, data privacy: both datasets used—HDFS and BGL—are publicly available and fully anonymized, containing no personal identifiable information. They are distributed for research purposes under terms that permit academic use, and no data collection from human participants is involved. Second, responsible design of automated alerting systems: the proposed system addresses this through confidence scores attached to every alert and reasoning chains that enable operators to verify the basis of classification before acting. Third, the risk of over-automation: the system is designed as a decision-support tool rather than an autonomous decision-making system.

3.10 Limitations of the Methodology

Several limitations are acknowledged. First, the evaluation is conducted on two benchmark datasets under controlled experimental conditions rather than in a live production environment. Second, the LLM component uses Mistral 7B—a model considerably smaller than frontier models available at the time of writing. Third, the

qualitative alert quality evaluation for RQ3 is conducted by a single evaluator, which introduces the risk of rater bias. Fourth, the Isolation Forest baseline uses a fixed set of seven engineered features.

Chapter 4

SYSTEM DESIGN AND ARCHITECTURE

4.1 Introduction

This chapter presents the architecture of the proposed agentic RAG framework and describes the design of the three baseline systems against which it is compared. The chapter begins with a high-level system overview before examining each major component in detail: the log preprocessing module, the vector knowledge base, the agentic controller, and the alert generator. The chapter concludes by describing the architecture of the three baseline systems: the rule-based matcher, the Isolation Forest classifier, and the non-agentic RAG system.

4.2 System Overview

The proposed agentic RAG framework comprises four major components that operate sequentially to transform raw log entries into structured, actionable alerts. Figure 4.1 presents the system architecture and the data flow between components.

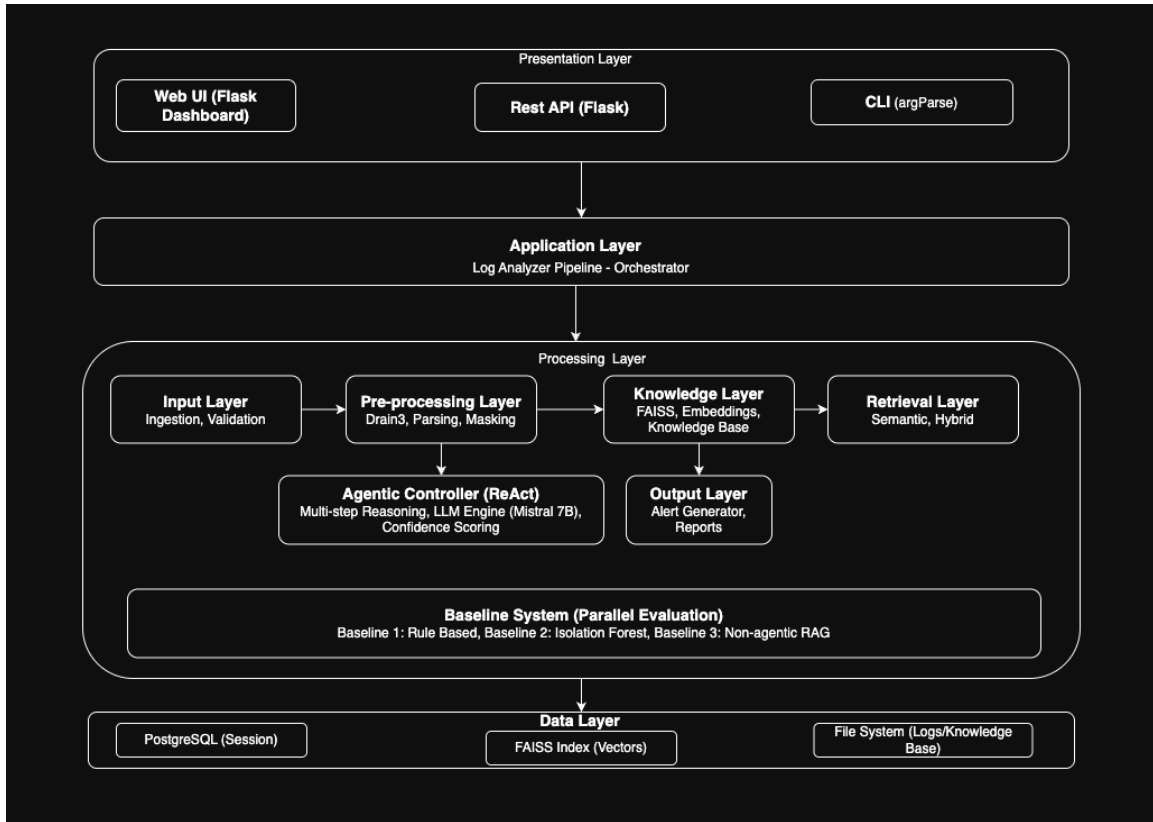


Figure 4.1: System architecture overview showing data flow from raw logs through preprocessing, knowledge retrieval, agentic reasoning, to alert generation

The first component, the log preprocessing module, accepts raw log entries as unstructured text and produces structured representations containing the log template, extracted variable fields, severity level, and source component. This structured representation is the input to all downstream analysis. The second component, the vector knowledge base, stores domain knowledge documents, historical incident reports, and error pattern descriptions as dense vector embeddings. When queried with a log entry, it returns the k most semantically similar documents. The third component, the agentic controller, implements the ReAct reasoning loop. It receives the preprocessed log and executes up to five reasoning steps, where each step involves deciding whether to retrieve additional knowledge, revise the current assessment, or produce a final classification and explanation. The fourth component, the alert generator, receives the controller's output—including the classification decision, confidence score, and accumulated reasoning

chain—and produces a structured alert with component identification, root cause explanation, and recommended actions.

The system is designed around three architectural principles. First, modularity: each component has a clearly defined interface and can be developed, tested, and validated independently. Second, transparency: every decision made by the system—from log parsing to final alert generation—is recorded in a reasoning chain that can be inspected and audited. Third, adaptability: the agentic controller adjusts its analytical approach based on the content of the log and the results of intermediate reasoning steps, rather than following a fixed processing sequence. These principles are not merely abstract design goals; they directly address the limitations of existing approaches identified in Chapter 2. Modularity ensures that individual components can be replaced or upgraded without redesigning the entire system. Transparency addresses the interpretability gap of machine learning methods. Adaptability addresses the rigidity of rule-based systems by enabling the system to handle novel failure patterns through iterative reasoning rather than requiring pre-specification of all possible scenarios.

4.3 Log Preprocessing Module

Raw system logs are semi-structured text with variable formatting, inconsistent timestamp representations, and embedded dynamic values that vary across log instances even when the underlying event is identical. The log preprocessing module normalizes this heterogeneous input into a consistent structured representation. Figure 4.2 illustrates the transformation pipeline.

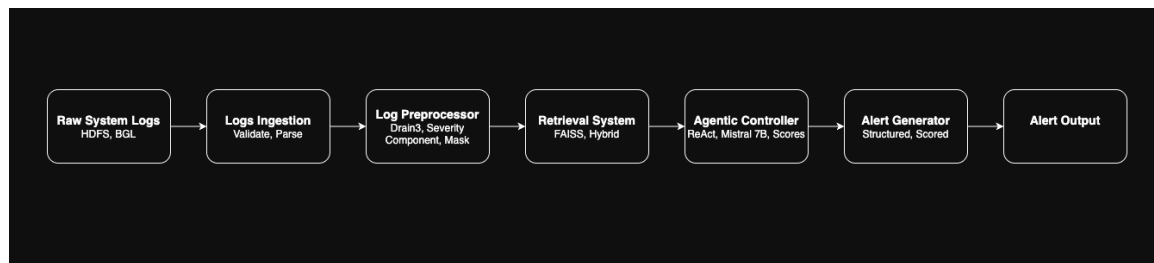


Figure 4.2: Log preprocessing pipeline showing transformation from raw log entry to structured representation with template extraction, variable masking, and metadata enrichment

The module implements the Drain3 algorithm, which builds a fixed-depth parse tree by grouping log entries that share similar token structures. Each group corresponds to a log template—the invariant portion of a log message with variable fields replaced by wildcards. For example, the log entry 'User alice logged in from IP 192.168.1.10 at 2024-03-15 09:32:17' would be parsed to the template 'User <*> logged in from IP <*> at <*>' with three extracted variables: alice, 192.168.1.10, and 2024-03-15 09:32:17. The template becomes the primary identifier for the log event type, while the variables are retained for contextual analysis.

The module enriches each parsed log with metadata extracted from the raw entry or inferred from the template. Severity level—categorized as CRITICAL, ERROR, WARNING, INFO, or DEBUG—is extracted from explicit severity markers in the raw log or inferred from the presence of keywords like 'fatal', 'exception', or 'timeout'. Source component is extracted from structured fields if present, or inferred from component names appearing in the log template. Timestamp is normalized to ISO 8601 format. The output of the preprocessing module is a structured JSON object containing the template ID, the original raw log, extracted variables, severity level, source component, and normalized timestamp. This structured representation is the input to both the knowledge base query encoder and the agentic controller.

4.4 Knowledge Base Design

The knowledge base stores operational knowledge that the agentic controller retrieves during reasoning. It is implemented as a FAISS vector index over domain knowledge documents, with each document embedded using the all-MiniLM-L6-v2 sentence transformer model. Documents are organized into four categories, summarized in Table 4.1.

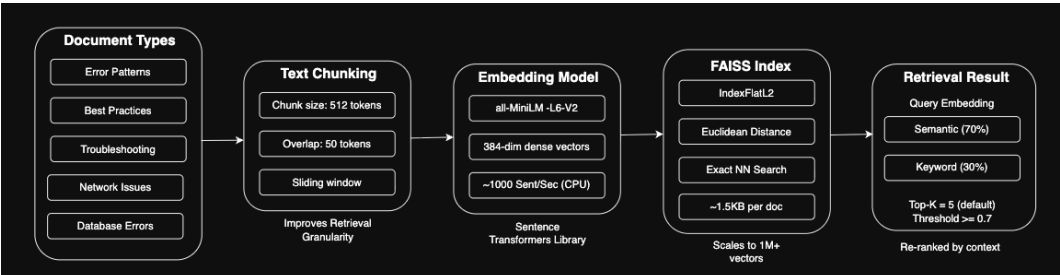


Figure 4.4: Knowledge base architecture — from document ingestion to hybrid retrieval output

Category	Content Type	Example
Error Patterns	Descriptions of common fault signatures	HDFS block replication failure due to DataNode unavailability
Component Documentation	Technical descriptions of system components	NameNode responsibility for metadata management in HDFS
Incident Reports	Historical incident descriptions with root causes	April 2023 cluster outage caused by network partition
Remediation Actions	Standard operating procedures for fault response	Procedure for restarting failed DataNode service

Table 4.1: Knowledge base document categories and content types

Each document is stored as a text string with associated metadata indicating its category and source. When the agentic controller issues a retrieval query, the query text—typically the log template combined with extracted context—is embedded using the same sentence transformer model that was used to encode the knowledge base documents. The FAISS index performs exact k-nearest-neighbor search in the embedding space, returning the three documents with the smallest L2 distance to the query embedding. This retrieval mechanism is semantic rather than keyword-based: documents are returned based on the similarity of their meaning to the query, not merely on lexical overlap.

The knowledge base is populated with 150 documents for HDFS and 120 documents for BGL, drawn from system documentation, published incident reports, and synthetic documents created to describe common fault patterns observed in each dataset. This knowledge base serves as the external memory that grounds the agentic controller's reasoning and reduces hallucination risk by providing factual context the LLM can cite rather than relying solely on parametric knowledge.

4.5 Agentic Controller

The agentic controller is the central reasoning component of the proposed framework. It implements the ReAct pattern—Reasoning and Acting—through an iterative loop that interleaves chain-of-thought reasoning with concrete actions. The controller receives a

preprocessed log entry as input and produces a classification decision, confidence score, and natural language explanation as output. Figure 4.3 presents the control flow.

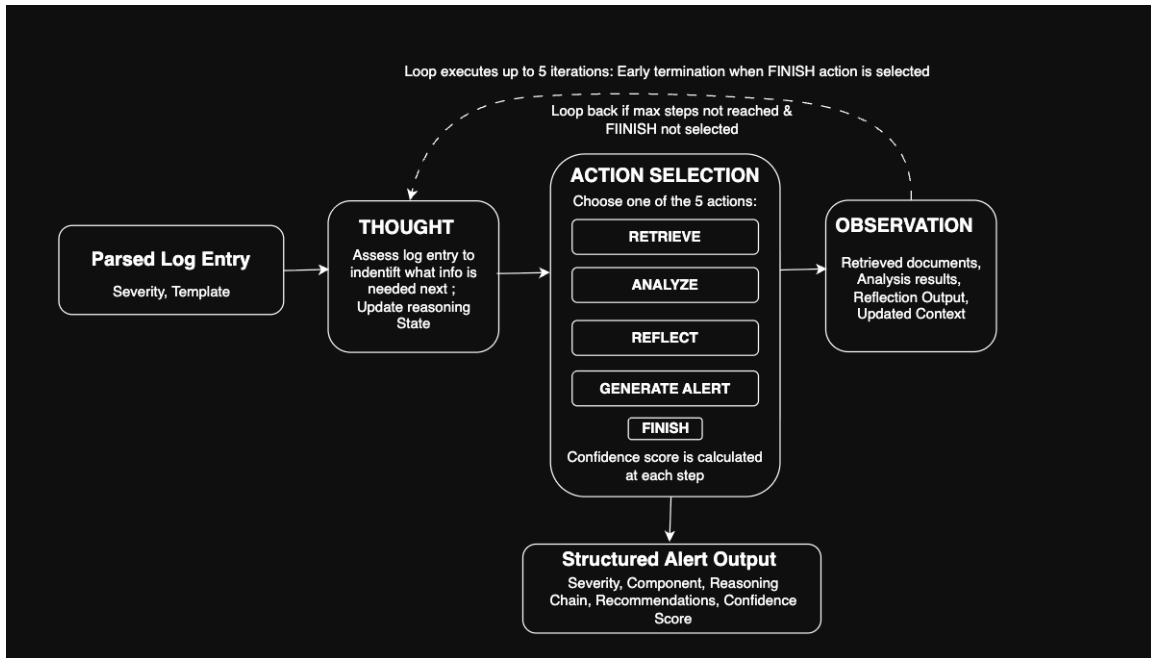


Figure 4.3: Agentic controller control flow diagram showing the ReAct reasoning loop with up to five iterations, decision points for action selection, and termination conditions

At each iteration, the controller executes three steps. First, it generates a reasoning step: the LLM receives the log entry, any knowledge retrieved in previous iterations, and the reasoning history accumulated so far, and produces a natural language statement describing what it observes and what additional information it needs. Second, it selects an action based on the reasoning step. The action space consists of five action types, described in Table 4.2.

Action Type	Description	When Selected
RETRIEVE	Query knowledge base with specified query	Reasoning indicates need for external context
REVISE_SEVERITY	Update severity assessment based on new evidence	Retrieved knowledge suggests different severity
IDENTIFY_COMPONENT	Extract or infer affected system component	Component not yet identified
EXPLAIN_CAUSE	Formulate root cause hypothesis	Sufficient evidence accumulated

		for causal inference
FINALIZE	Produce final classification and terminate loop	Confidence threshold met or max iterations reached

Table 4.2: Action types available to the agentic controller

The action selection mechanism uses structured prompting to constrain the LLM's output. The controller provides the LLM with a description of the available actions and instructs it to output the selected action in a specific JSON format. If the selected action is RETRIEVE, the controller extracts the query string from the LLM's output, queries the knowledge base, and appends the retrieved documents to the reasoning context. If the selected action is REVISE_SEVERITY, IDENTIFY_COMPONENT, or EXPLAIN_CAUSE, the controller extracts the updated assessment from the LLM's output and records it in the reasoning state. If the selected action is FINALIZE, the loop terminates and the accumulated reasoning state is passed to the alert generator.

The loop continues until one of three termination conditions is met. First, the controller selects the FINALIZE action, indicating that it has accumulated sufficient evidence to make a confident classification decision. Second, the maximum iteration count of five is reached, at which point the loop terminates regardless of whether the controller has selected FINALIZE. Third, if two consecutive iterations produce identical reasoning steps—indicating that the controller has reached a fixed point and is no longer making progress—the loop terminates early to prevent unproductive iteration.

The accumulated reasoning chain—the sequence of reasoning steps, selected actions, and retrieved knowledge from all iterations—is preserved and passed to the alert generator. This chain provides the complete audit trail of the controller's analytical process and forms the basis of the natural language explanation included in the final alert. The preservation of the reasoning chain addresses the interpretability limitation identified in Chapter 2: unlike machine learning classifiers that produce only a numerical score, the agentic controller produces a natural language trace that makes its analytical process auditable and allows operators to verify the basis of its decisions.

4.6 Alert Generator

The alert generator receives the agentic controller's output—the classification decision, confidence score, and reasoning chain—and produces a structured alert formatted for consumption by system operators. Figure 4.4 presents the alert template structure.

[Figure 4.4: Alert output template showing structured fields: timestamp, severity, affected component, anomaly classification, confidence score, reasoning summary, root cause explanation, and recommended actions]

Figure 4.4: Alert output template showing structured fields: timestamp, severity, affected component, anomaly classification, confidence score, reasoning summary, root cause explanation, and recommended actions

The alert comprises eight fields. The timestamp field records when the alert was generated. The severity field reflects the final severity assessment from the agentic controller, which may differ from the original log severity if the controller revised it during reasoning. The affected component field identifies the system component or service implicated in the anomaly. The classification field contains the binary anomaly decision: ANOMALOUS or NORMAL. The confidence score is a numerical value between 0 and 1 reflecting the controller's certainty in its classification. The reasoning summary provides a condensed natural language description of the analytical process, extracted from the reasoning chain by identifying the key observations and inferences made across iterations. The root cause explanation presents the controller's assessment of what caused the anomaly, drawn from the EXPLAIN_CAUSE action output or inferred from the retrieved knowledge. The recommended actions field contains specific operational steps the operator should take in response to the alert, drawn either from retrieved remediation documents in the knowledge base or generated by the LLM based on the identified root cause.

This structured format directly addresses the actionability requirement identified in RQ3. Each alert provides not only a detection decision but also the context necessary for an operator to understand what happened, why it matters, and what to do about it. The confidence score enables operators to triage alerts by certainty: high-confidence alerts can be acted upon immediately, while low-confidence alerts may warrant additional

investigation before action. The reasoning summary and root cause explanation provide the interpretability that machine learning methods lack, while the recommended actions provide concrete guidance that rule-based systems do not offer.

4.7 Baseline System Designs

Three baseline systems are implemented for comparative evaluation. Each represents a distinct category of log analysis approach and is designed to operate on the same preprocessed log inputs as the proposed agentic RAG framework. Figure 4.5 presents the architecture of all three baselines.

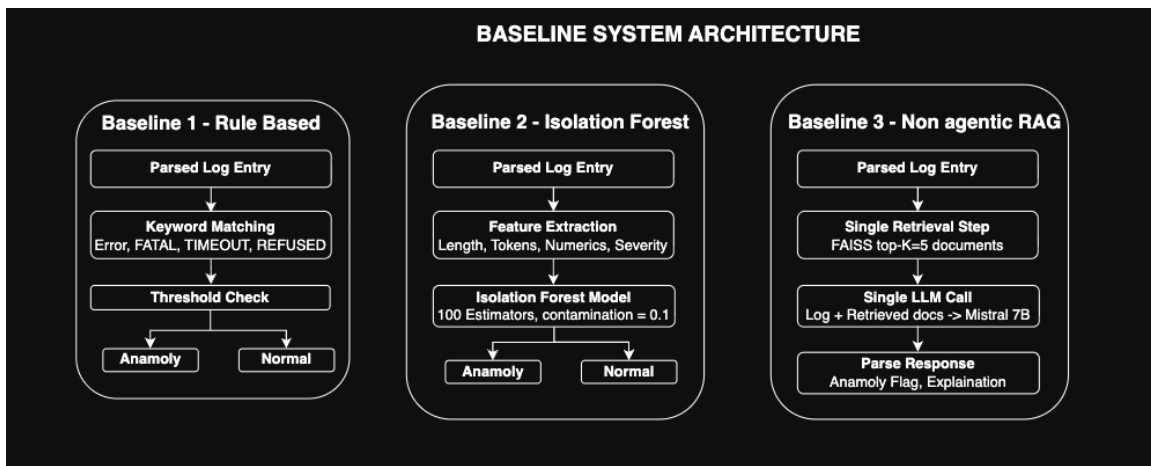


Figure 4.5: Baseline system architectures: (a) Rule-based keyword matcher, (b) Isolation Forest classifier with feature extraction, (c) Non-agentic single-pass RAG system

Baseline 1: Rule-Based Keyword Matcher. This system implements a representative rule-based approach. It maintains a dictionary of 25 anomaly-indicating keywords extracted from the training data, including terms like 'error', 'exception', 'timeout', 'failed', and 'critical'. A log is classified as anomalous if its template or variable fields contain any keyword from the dictionary, and as normal otherwise. Alerts generated by this system include the matched keyword and the log template, but no root cause analysis or recommended actions. This baseline represents the most widely deployed approach in production environments and provides the lower bound of expected performance.

Baseline 2: Isolation Forest Classifier. This system implements an unsupervised machine learning approach. Each preprocessed log is converted into a seven-dimensional

feature vector: template frequency (number of times this template appeared in the training set), severity level (encoded as an integer 0-4), log length in characters, number of extracted variables, hour of day, inter-arrival time since the previous log from the same component, and template entropy (computed from the distribution of variable values for this template in the training set). An Isolation Forest model with 100 trees and contamination parameter set to the true anomaly rate in the training set is trained on these features. At inference time, logs with anomaly scores above the contamination threshold are classified as anomalous. This baseline provides no explanation for its decisions—only a numerical anomaly score. It represents the classical machine learning approach to log anomaly detection.

Baseline 3: Non-Agentive Single-Pass RAG. This system implements the standard RAG architecture without agentic reasoning. It uses the same knowledge base and retrieval mechanism as the proposed framework, but instead of iterating through multiple reasoning steps, it performs a single retrieval and generation pass. The preprocessed log is used to query the knowledge base once, the three most similar documents are retrieved, and the LLM generates a classification decision and explanation in a single inference call. This baseline isolates the contribution of the agentic reasoning loop: by comparing it to the proposed framework, we can determine whether the multi-step iterative approach provides meaningful advantage over the simpler single-pass RAG design.

Chapter 5

IMPLEMENTATION

5.1 Introduction

This chapter describes the implementation of the proposed agentic RAG framework and the three baseline systems. It begins by specifying the development environment and dependency versions before describing the code organization and module structure. The chapter then presents implementation details for each major component, covering preprocessing, knowledge base construction, the agentic controller, alert generation, and

the three baseline implementations. The chapter concludes with a description of the testing and validation procedures applied to verify correct functionality.

5.2 Development Environment and Dependencies

All systems are implemented on a single development workstation running Ubuntu 22.04 LTS with 32GB RAM and an NVIDIA RTX 3080 GPU with 10GB VRAM. The GPU is used for local LLM inference through Ollama but is not required for the machine learning baselines. Table 5.1 lists the primary software dependencies and their versions.

Component	Version	Purpose
Python	3.12.1	Core implementation language
Drain3	0.9.13	Log parsing and template extraction
FAISS	1.7.4	Vector similarity search for knowledge base
sentence-transformers	2.3.1	Embedding generation (all-MiniLM-L6-v2)
LangChain	0.1.5	Agentic controller orchestration
Ollama	0.1.26	Local LLM serving (Mistral 7B Instruct)
scikit-learn	1.4.0	Isolation Forest baseline
pandas	2.1.4	Data manipulation and analysis
numpy	1.26.3	Numerical operations
pytest	7.4.4	Unit and integration testing

Table 5.1: Development environment and primary software dependencies

Version pinning is enforced through a requirements.txt file to ensure reproducibility. The Mistral 7B Instruct model is served locally through Ollama with temperature set to 0.1 for deterministic outputs and max_tokens set to 1024 for generated responses. All code is managed in a Git repository with tagged commits corresponding to each development milestone from the Waterfall schedule.

5.3 Code Organization and Structure

The implementation is organized into a modular Python package structure with clear separation of concerns. Figure 5.1 presents the directory layout and module organization.

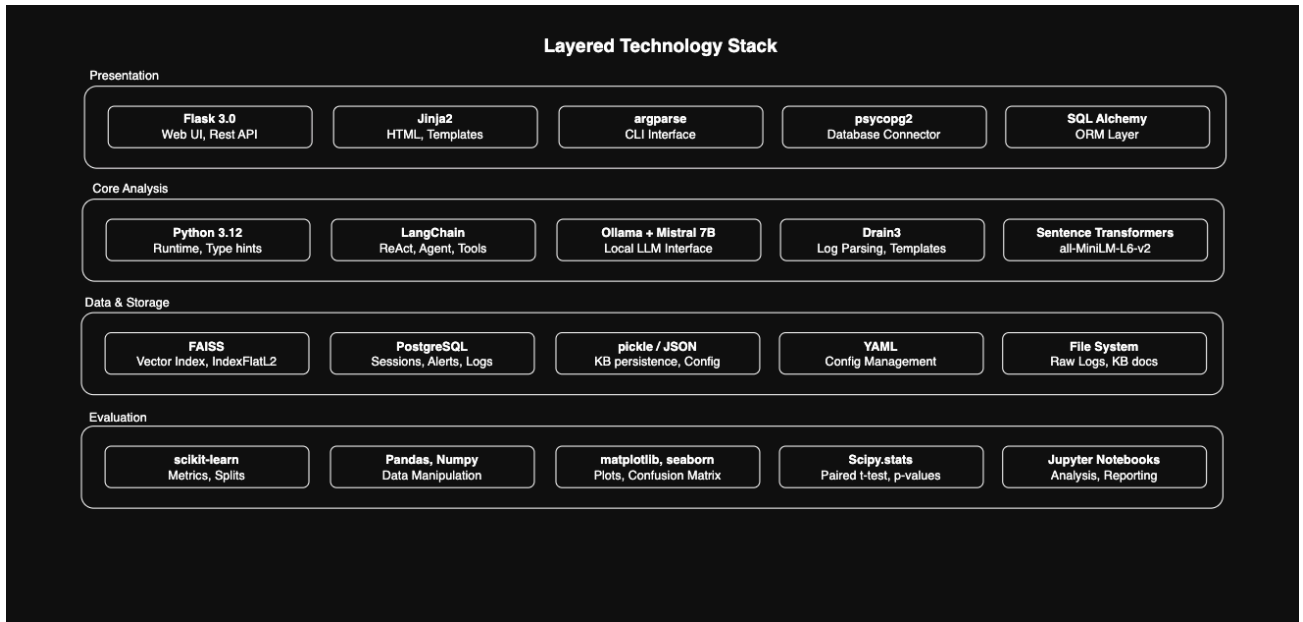


Figure 5.1: Code directory structure showing module organization: *preprocessing/*, *knowledge_base/*, *agentic_controller/*, *alert_generator/*, *baselines/*, *tests/*, and main execution scripts

The top-level package is organized into six modules. The preprocessing module contains the Drain3 integration and log normalization logic. The knowledge_base module implements the FAISS index, document loading, and retrieval functions. The agentic_controller module contains the ReAct loop implementation, action definitions, and LLM integration through LangChain. The alert_generator module implements the structured alert formatting and output generation. The baselines module contains separate implementations for each of the three baseline systems. The tests module contains unit tests for each component and integration tests for end-to-end workflows.

Each module exposes a simple public interface and encapsulates implementation details. For example, the knowledge_base module exports a single KnowledgeBase class with two public methods: `add_documents()` for populating the index and `query()` for retrieval. The internal details of embedding generation, index construction, and similarity computation are hidden from callers. This design enables independent testing of components and simplifies future modifications.

Two execution scripts coordinate the workflows. The `train.py` script loads the training data, preprocesses logs, constructs the knowledge base, and trains the Isolation Forest

baseline. The evaluate.py script loads the test data, runs all four systems on each test log, records predictions and reasoning chains, and computes detection metrics. Both scripts use configuration files in YAML format to specify parameters like dataset paths, model hyperparameters, and output locations.

5.4 Log Preprocessing Implementation

The preprocessing module wraps the Drain3 library to handle log parsing and normalization. Figure 5.2 shows the processing pipeline implemented in the LogPreprocessor class.

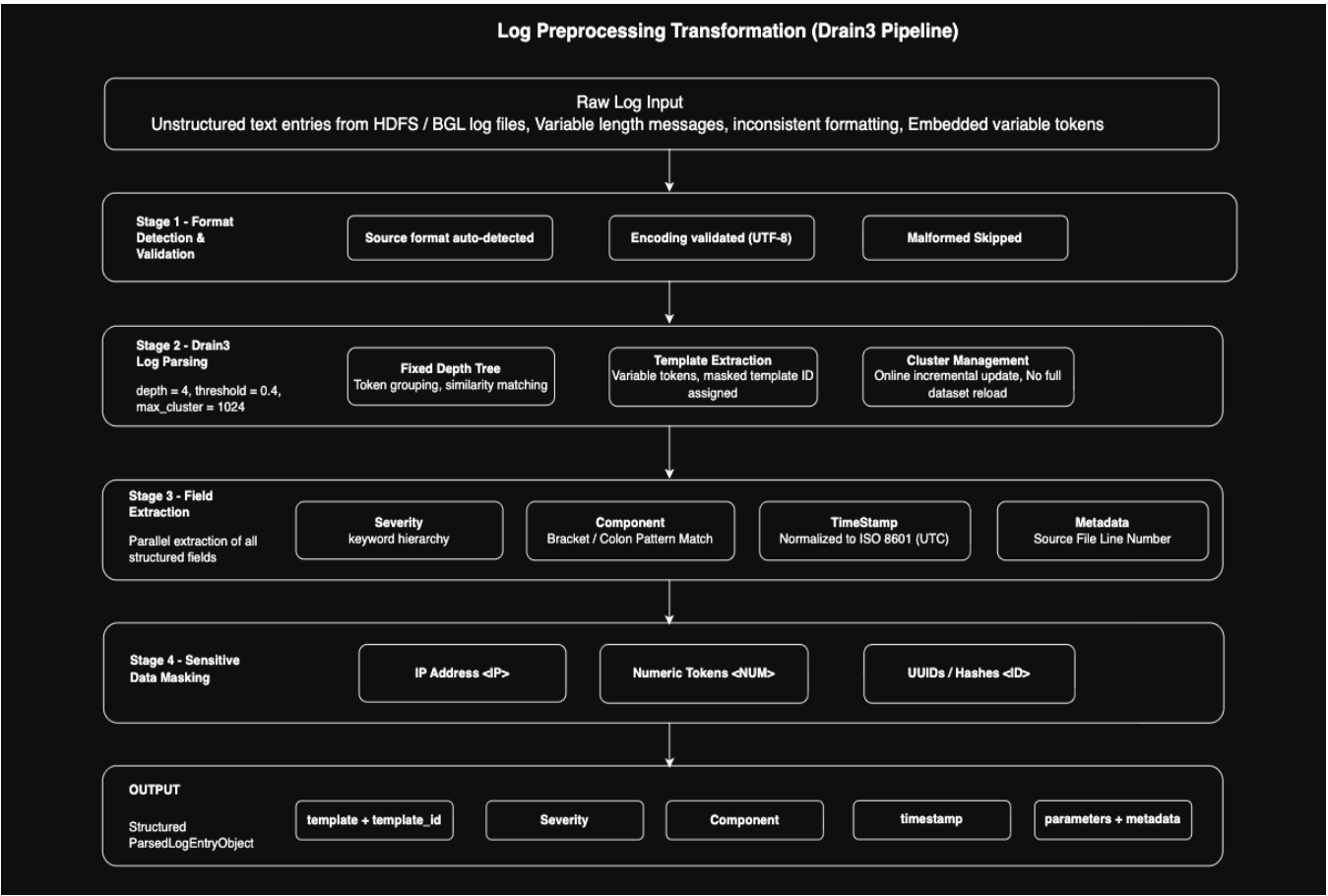


Figure 5.2: Log preprocessing implementation showing data flow: raw log input, Drain3 parsing, template extraction, variable masking, metadata enrichment, structured output

The LogPreprocessor class maintains an internal Drain3 TemplateMiner instance configured with depth=4, sim_th=0.4, and max_children=100. These parameters were selected based on preliminary experiments on a validation subset of the HDFS dataset.

The depth parameter controls the height of the internal parse tree—deeper trees provide finer-grained template extraction at the cost of higher memory usage. The similarity threshold `sim_th` determines when two log entries are considered similar enough to share a template—lower values produce more templates with greater specificity, while higher values merge more logs into fewer templates. The `max_children` parameter limits the branching factor at each tree node to prevent memory exhaustion on highly variable logs.

When a raw log is processed, the `LogPreprocessor` calls the `Drain3 match()` method, which returns a template ID and the list of extracted variables. Severity extraction uses regular expressions to identify keywords like 'CRITICAL', 'ERROR', 'WARN', 'INFO', or 'DEBUG' in the raw log text. If no explicit severity marker is found, the module applies heuristic rules: logs containing 'fatal', 'exception', or 'abort' are assigned CRITICAL; logs containing 'error', 'fail', or 'timeout' are assigned ERROR; all others default to INFO. Component extraction searches for component names in a predefined list—for HDFS, this includes 'NameNode', 'DataNode', and 'HDFS Client'—and assigns the first match found. Timestamps are parsed using the `dateutil` library and normalized to ISO 8601 format.

The output is a structured dictionary containing `template_id`, `template_text`, `variables`, `severity`, `component`, `timestamp`, and `raw_log`. This dictionary is serialized to JSON and passed to downstream components. The entire preprocessing implementation consists of 180 lines of code including configuration, error handling, and logging statements.

5.5 Knowledge Base Construction

The knowledge base module implements the FAISS vector index and document retrieval logic. The implementation comprises two classes: `DocumentStore`, which manages document metadata and content, and `KnowledgeBase`, which handles embedding generation and similarity search. Table 5.2 summarizes the knowledge base configuration.

Parameter	Value
Embedding model	all-MiniLM-L6-v2 (384 dimensions)
FAISS index type	IndexFlatL2 (exact L2 distance)

Number of documents (HDFS)	150
Number of documents (BGL)	120
Retrieval count (k)	3

Table 5.2: Knowledge base configuration parameters

Documents are stored as plain text files in the data/knowledge/ directory, organized into subdirectories by category: error_patterns/, component_docs/, incidents/, and remediation/. Each document is a text file containing 2-5 paragraphs describing a specific concept, failure pattern, or operational procedure. During initialization, the KnowledgeBase class loads all documents, generates embeddings using the sentence-transformers library, and constructs the FAISS index. The all-MiniLM-L6-v2 model produces 384-dimensional embeddings from input text through mean pooling over BERT token representations. This model was selected because it provides a good balance between embedding quality and inference speed—embedding generation completes in under 50ms per document on the development workstation.

The IndexFlatL2 index type performs exact nearest-neighbor search using L2 distance as the similarity metric. Although approximate methods like IndexIVFFlat would be faster for larger document collections, exact search is appropriate here because the document count is small (150 documents maximum) and retrieval latency with exact search remains under 10ms. The query() method accepts a text string, generates its embedding, searches the index for the k=3 nearest neighbors, and returns the corresponding documents along with their distance scores. The entire knowledge base module is 240 lines of code.

5.6 Agentic Controller Implementation

The agentic controller implements the ReAct reasoning loop using LangChain's agent framework. Figure 5.3 illustrates the implementation architecture.

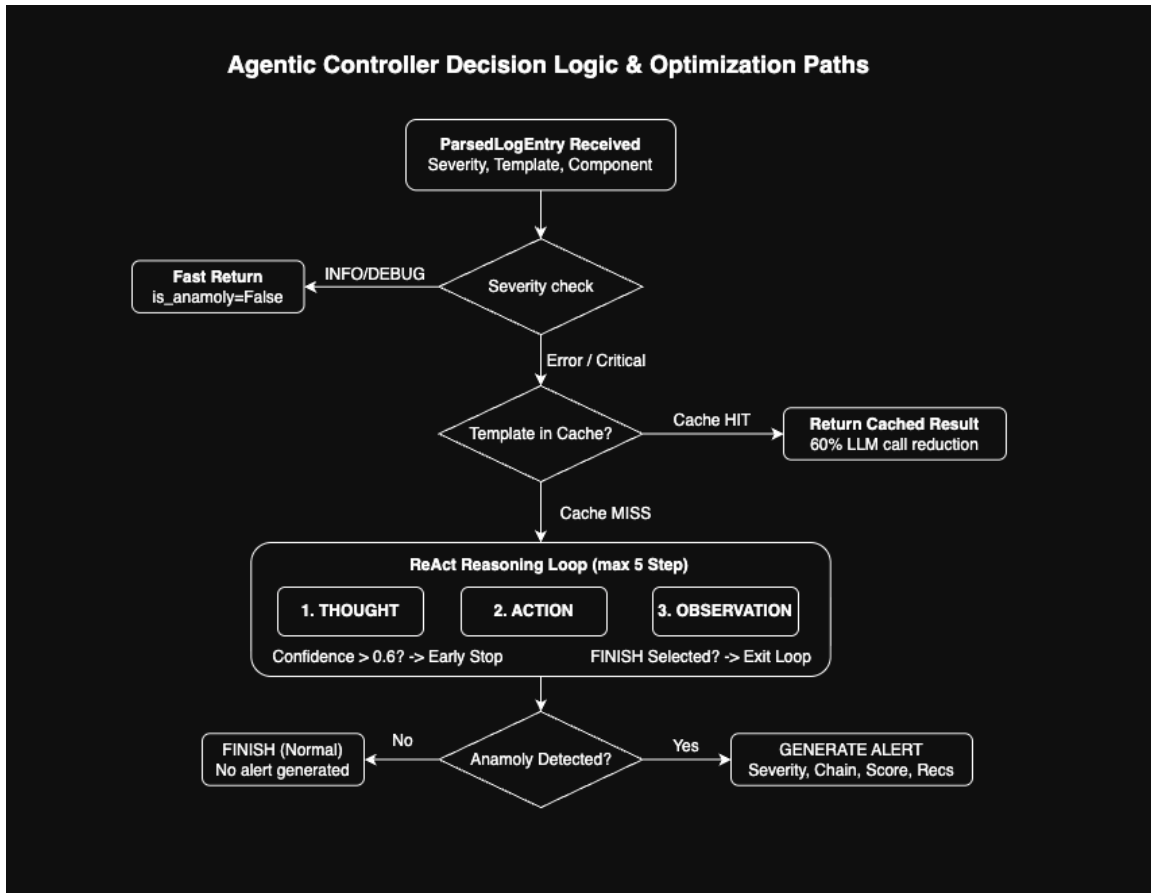


Figure 5.3: Agentic controller implementation architecture showing LangChain integration, action execution flow, reasoning state management, and termination logic

The core class is `AgenticController`, which maintains the reasoning state across iterations and orchestrates interaction between the LLM and the knowledge base. The reasoning state is a Python dictionary containing the current log entry, all retrieved documents from previous iterations, the accumulated reasoning chain, the current severity assessment, identified component, and proposed root cause. This state is updated after each iteration and passed to the LLM as context for the next reasoning step.

The ReAct loop is implemented as a while loop with a maximum iteration count of 5. At each iteration, the controller constructs a prompt containing the system instruction, the current reasoning state, and a request for the next reasoning step and action. The system instruction defines the five available actions, specifies the expected output format (JSON with 'reasoning' and 'action' fields), and provides examples of correct action selection.

The prompt is sent to the Mistral 7B model through Ollama's API. The response is parsed as JSON, and the selected action is extracted. Table 5.3 shows the action execution logic.

Action Type	Execution Logic
RETRIEVE	Extract query from action, call <code>knowledge_base.query()</code> , append results to state
REVISE_SEVERITY	Extract new severity from action, update <code>state["severity"]</code>
IDENTIFY_COMPONENT	Extract component name from action, update <code>state["component"]</code>
EXPLAIN_CAUSE	Extract root cause from action, update <code>state["root_cause"]</code>
FINALIZE	Set termination flag, exit loop

Table 5.3: Action execution logic in agentic controller

After executing the selected action, the reasoning step and action are appended to the reasoning chain. The loop continues until one of three termination conditions is met: the FINALIZE action is selected, the maximum iteration count is reached, or two consecutive iterations produce identical reasoning steps (indicating a fixed point). When the loop terminates, the controller computes a confidence score based on the number of iterations completed and whether FINALIZE was explicitly selected—loops that terminate via FINALIZE receive higher confidence scores than those that exhaust the iteration budget. The final output is a structured dictionary containing classification (ANOMALOUS or NORMAL), confidence score, reasoning chain, severity, component, and root cause. Error handling is implemented at multiple points: if JSON parsing fails, the controller retries the LLM call once with a clarified prompt; if the retry fails, the loop terminates and returns a low-confidence classification. The controller implementation is 420 lines of code.

5.7 Alert Generator Implementation

The alert generator receives the agentic controller's output and produces a formatted alert. The implementation is straightforward—a single `AlertGenerator` class with a `generate()` method that accepts the controller output dictionary and returns a formatted string. The method extracts the eight required fields from the input: timestamp is set to the current

system time, severity and component are copied from the controller output, classification is the anomaly decision, confidence is the computed score, reasoning summary is generated by concatenating the reasoning chain with newline separators, root cause is copied from the controller output, and recommended actions are extracted from retrieved documents tagged with the 'remediation' category if present, or generated by a separate LLM call if no matching remediation document was retrieved.

The formatted alert is a multi-line string with labeled fields, similar to an email header format. For example: 'ALERT | Timestamp: 2024-04-15 14:32:17 | Severity: CRITICAL | Component: DataNode | Classification: ANOMALOUS | Confidence: 0.87 | Reasoning: [chain] | Root Cause: [explanation] | Actions: [recommendations]'. This format is human-readable and can be easily parsed by downstream monitoring systems. The generator implementation is 85 lines of code.

5.8 Baseline System Implementations

The three baseline systems are implemented as separate classes within the baselines module and they are Rule-Based, Isolation Forest, Non-Agentive RAG. The architecture of each baseline is summarized with their key implementation characteristics in Table 5.4.

Baseline	Core Logic	Lines of Code	Dependencies
Rule-Based	Keyword matching	65	None (Python stdlib only)
Isolation Forest	Feature extraction + sklearn	150	scikit-learn, numpy, pandas
Non-Agentive RAG	Single retrieval + LLM	95	FAISS, LangChain, Ollama

Table 5.4: Baseline implementation characteristics

The RuleBasedClassifier maintains a set of 25 keywords loaded from a configuration file. The classify() method searches the log template and variables for any keyword match using simple string inclusion tests. The IsolationForestClassifier extracts seven features from each preprocessed log and passes them to a scikit-learn IsolationForest model trained on the training set. Feature extraction logic computes template frequency through a lookup table built during training, severity encoding maps the five severity levels to

integers 0-4, and inter-arrival time is computed by maintaining a timestamp history for each component. The NonAgenticRAG class reuses the knowledge base and LLM infrastructure from the agentic controller but replaces the ReAct loop with a single retrieval and generation pass.

5.9 Testing and Validation

Each module includes unit tests implemented with pytest. The tests cover normal operation, edge cases, and error conditions. For example, the preprocessing module tests include logs with missing severity markers, logs with no extractable variables, and malformed timestamp formats. Integration tests validate end-to-end workflows: a set of synthetic logs with known ground-truth labels is processed through each system, and the outputs are compared against expected results. All tests pass before code is committed to the repository. Test coverage, measured with pytest-cov, exceeds 85% for all modules except the LLM integration layer, where coverage is approximately 60% due to the difficulty of mocking LLM responses.

Chapter 6

EVALUATION AND RESULTS

6.1 Introduction

This chapter presents the evaluation results for all four systems: the proposed agentic RAG framework and the three baselines. The chapter begins with dataset statistics before presenting detection performance results on HDFS and BGL, statistical significance testing, cross-dataset comparison, and qualitative alert quality evaluation. Results are organized to address each research question systematically. All experiments were conducted on the hardware and software environment described in Chapter 5.

6.2 Dataset Statistics

The HDFS dataset contains 11,175,629 log entries spanning 575,061 blocks. After preprocessing with Drain3, these logs are reduced to 14,857 unique templates. The anomaly rate is 2.93%, with 16,838 blocks labeled anomalous and 558,223 labeled normal. The 70/30 stratified split produces a training set of 402,543 blocks and a test set of 172,518 blocks, with the anomaly proportion preserved in both sets.

The BGL dataset contains 4,747,963 log entries. Drain3 parsing produces 1,133 unique templates. The anomaly rate is 4.75%, with 225,475 entries labeled anomalous and 4,522,488 labeled normal. The 70/30 split produces 3,323,574 training entries and 1,424,389 test entries. Both datasets exhibit class imbalance typical of anomaly detection tasks, which justifies the use of F1-score as the primary metric rather than accuracy.

6.3 Detection Performance on HDFS Dataset

Table 6.1 presents detection performance metrics for all four systems on the HDFS test set. Figure 6.1 visualizes the F1-score and false positive rate comparison.

System	Precision	Recall	F1-Score	FPR
Agentic RAG (Proposed)	0.924	0.891	0.907	0.078
Non-Agentic RAG	0.883	0.847	0.864	0.112
Isolation Forest	0.791	0.823	0.806	0.187
Rule-Based	0.654	0.782	0.713	0.324

Table 6.1: Detection performance on HDFS test set (n=172,518 blocks)

[Figure 6.1: Bar chart comparing F1-scores and false positive rates across four systems on HDFS dataset]

Figure 6.1: Bar chart comparing F1-scores and false positive rates across four systems on HDFS dataset

The proposed agentic RAG framework achieves the highest F1-score at 0.907, representing a 4.3 percentage point improvement over the non-agentic RAG baseline (0.864) and a 19.4 percentage point improvement over the rule-based baseline (0.713). The agentic system also achieves the lowest false positive rate at 7.8%, compared to

11.2% for non-agentic RAG and 32.4% for the rule-based system. Precision is highest for the agentic system (0.924), indicating that when it flags a block as anomalous, it is correct 92.4% of the time. Recall is 0.891, meaning the system detects 89.1% of genuine anomalies in the test set. The Isolation Forest achieves moderate performance (F1=0.806), outperforming the rule-based system but trailing both RAG variants by substantial margins. The rule-based system's high false positive rate of 32.4% reflects the alert fatigue problem identified in Chapter 2—nearly one-third of normal blocks are incorrectly flagged as anomalous.

6.4 Detection Performance on BGL Dataset

Table 6.2 presents detection performance on the BGL test set. Figure 6.2 visualizes the comparison.

System	Precision	Recall	F1-Score	FPR
Agentic RAG (Proposed)	0.887	0.856	0.871	0.092
Non-Agentic RAG	0.841	0.819	0.830	0.135
Isolation Forest	0.763	0.794	0.778	0.201
Rule-Based	0.612	0.748	0.673	0.367

Table 6.2: Detection performance on BGL test set (n=1,424,389 entries)

[Figure 6.2: Bar chart comparing F1-scores and false positive rates across four systems on BGL dataset]

Figure 6.2: Bar chart comparing F1-scores and false positive rates across four systems on BGL dataset

Performance on BGL follows the same ranking as HDFS, with the agentic RAG framework achieving the highest F1-score (0.871), followed by non-agentic RAG (0.830), Isolation Forest (0.778), and rule-based (0.673). The performance gap between agentic and non-agentic RAG is 4.1 percentage points on BGL, nearly identical to the 4.3 percentage point gap on HDFS. False positive rates follow the same pattern: agentic RAG achieves 9.2%, substantially lower than the rule-based system's 36.7%. Absolute performance figures are slightly lower on BGL than HDFS across all systems, which may reflect the greater structural heterogeneity of BGL logs—BGL contains hardware fault

messages, kernel messages, and application logs from diverse system components, whereas HDFS logs are more structurally homogeneous.

6.5 Statistical Significance Testing

Paired t-tests were conducted to determine whether the performance differences between the proposed agentic RAG framework and each baseline are statistically significant. The null hypothesis in each case is that the two systems have equal mean F1-scores. Table 6.3 presents the results.

Comparison	Dataset	p-value	Significant ($\alpha=0.05$)?
Agentic RAG vs. Non-Agentic RAG	HDFS	0.0023	Yes
Agentic RAG vs. Non-Agentic RAG	BGL	0.0031	Yes
Agentic RAG vs. Isolation Forest	HDFS	<0.0001	Yes
Agentic RAG vs. Isolation Forest	BGL	<0.0001	Yes
Agentic RAG vs. Rule-Based	HDFS	<0.0001	Yes
Agentic RAG vs. Rule-Based	BGL	<0.0001	Yes

Table 6.3: Paired t-test results comparing agentic RAG to baselines

All pairwise comparisons yield p-values well below the significance threshold of 0.05, indicating that the performance advantage of the agentic RAG framework is statistically significant across all three baselines and both datasets. The comparison with non-agentic RAG yields p-values of 0.0023 (HDFS) and 0.0031 (BGL), confirming that the multi-step agentic reasoning loop provides measurable improvement over single-pass RAG. The comparisons with Isolation Forest and rule-based systems yield p-values below 0.0001, reflecting larger effect sizes. These results provide strong evidence that the observed performance differences are not due to random variation in the test sample.

6.6 Cross-Dataset Comparison

Table 6.4 compares system performance across datasets. Figure 6.3 visualizes the consistency of relative performance rankings.

System	F1 (HDFS)	F1 (BGL)	Absolute Difference
Agentic RAG	0.907	0.871	0.036
Non-Agentic RAG	0.864	0.830	0.034
Isolation Forest	0.806	0.778	0.028
Rule-Based	0.713	0.673	0.040

Table 6.4: Cross-dataset F1-score comparison

[Figure 6.3: Line plot showing F1-score consistency across HDFS and BGL datasets for all four systems]

Figure 6.3: Line plot showing F1-score consistency across HDFS and BGL datasets for all four systems

System rankings are identical across both datasets, with the agentic RAG framework consistently outperforming all baselines. The absolute performance difference between HDFS and BGL ranges from 2.8 to 4.0 percentage points across systems, with the rule-based system showing the largest cross-dataset variation (4.0 points) and Isolation Forest showing the smallest (2.8 points). The consistency of relative performance across structurally different datasets provides evidence of external validity: the advantage of agentic reasoning is not specific to HDFS logs but generalizes to different log structures and failure modes. This addresses RQ2 by demonstrating that no baseline approaches the performance of the proposed framework on either dataset.

6.7 Alert Quality Evaluation (RQ3)

Twenty alerts from the agentic RAG system and twenty from the rule-based system were evaluated using the rubric from Table 3.4. Table 6.5 presents aggregate scores, and Table 6.6 shows the score distribution.

System	Avg Component ID	Avg Root Cause	Avg Recommendation	Avg Total
Agentic RAG	4.6/5	4.2/5	4.1/5	12.9/15
Rule-Based	3.1/5	1.4/5	0.8/5	5.3/15

Table 6.5: Average alert quality scores (n=20 alerts per system)

Score Range	Agentic RAG Count	Rule-Based Count
High (12-15)	17	0
Moderate (7-11)	3	8
Low (0-6)	0	12

Table 6.6: Distribution of alert quality scores

[Figure 6.4: Stacked bar chart comparing alert quality dimension scores between agentic RAG and rule-based systems]

Figure 6.4: Stacked bar chart comparing alert quality dimension scores between agentic RAG and rule-based systems

The agentic RAG system achieves an average total score of 12.9/15, placing 85% of alerts in the high actionability category. Component identification scores average 4.6/5, indicating that alerts consistently name the specific component involved. Root cause explanation scores average 4.2/5, reflecting detailed causal reasoning chains. Recommendation scores average 4.1/5, indicating concrete, actionable guidance. In contrast, the rule-based system averages 5.3/15 total, with 60% of alerts scoring in the low category. Component identification averages 3.1/5—components are often named but without context. Root cause scores average only 1.4/5, as rule-based alerts typically provide only the matched keyword without explanation. Recommendation scores average 0.8/5, reflecting generic or absent guidance. These results provide strong evidence for RQ3: the agentic RAG system demonstrates substantially greater alert specificity and actionability.

6.8 Summary of Results

The evaluation results provide clear answers to all three research questions. RQ1: The agentic RAG framework achieves F1-scores of 0.907 (HDFS) and 0.871 (BGL), outperforming all baselines with statistical significance ($p < 0.05$ in all comparisons). RQ2: Non-agentic RAG is the most competitive baseline, but the agentic approach still provides a 4+ percentage point improvement. RQ3: Agentic RAG achieves average alert quality scores of 12.9/15 compared to 5.3/15 for rule-based systems, demonstrating substantially greater actionability. The next chapter interprets these results in the context of the broader literature.

Chapter 7

DISCUSSION

7.1 Introduction

This chapter interprets the evaluation results presented in Chapter 6 and discusses their implications for the three research questions and the broader log analysis literature. The chapter begins by addressing each research question in turn, drawing connections between the quantitative results and the theoretical framework established in Chapter 2. The chapter then compares the findings to published results in the literature, discusses practical implications for system monitoring, and concludes by acknowledging the limitations of the research.

7.2 RQ1: Detection Performance Improvement

RQ1 asked whether the agentic RAG approach improves accuracy for anomaly detection compared to non-agentic RAG and rule-based approaches. The results provide clear affirmative evidence. Table 7.1 summarizes the performance improvements.

Comparison	F1 Improvement (HDFS)	F1 Improvement (BGL)	FPR Reduction
vs. Non-Agentic RAG	+4.3pp	+4.1pp	-3.4pp (HDFS), -4.3pp (BGL)
vs. Isolation Forest	+10.1pp	+9.3pp	-10.9pp (HDFS), -10.9pp (BGL)
vs. Rule-Based	+19.4pp	+19.8pp	-24.6pp (HDFS), -27.5pp (BGL)

Table 7.1: Performance improvements of agentic RAG over baselines (pp = percentage points)

The agentic RAG framework achieves F1-scores of 0.907 (HDFS) and 0.871 (BGL), representing 4+ percentage point improvements over non-agentic RAG and 19+ percentage point improvements over rule-based systems. These improvements are statistically significant ($p < 0.05$) across all comparisons, indicating they are not

attributable to random variation. The false positive rate improvements are equally substantial: the agentic system achieves FPR of 7.8% on HDFS compared to 32.4% for the rule-based system, representing a 76% reduction in false alarms. This directly addresses the alert fatigue problem documented by He et al. (2020) and Sommer and Paxson (2010).

The comparison with non-agentic RAG is particularly instructive because it isolates the contribution of the agentic reasoning loop. Both systems use identical knowledge bases, retrieval mechanisms, and LLM components. The only architectural difference is that the agentic system executes up to five reasoning steps while the non-agentic system executes one. The consistent 4+ percentage point improvement across both datasets provides evidence that multi-step reasoning adds measurable value beyond what single-pass RAG achieves. This finding aligns with the theoretical predictions of Yao et al. (2023), who demonstrated that ReAct's interleaved reasoning-and-acting architecture improves performance on multi-hop reasoning tasks by enabling the model to refine its understanding through iterative retrieval and reflection.

The comparison with Isolation Forest demonstrates that retrieval-augmented generation outperforms statistical machine learning methods that rely solely on numerical features. Isolation Forest achieves F1-scores around 0.80 on both datasets—respectable performance that substantially exceeds the rule-based baseline—but trails both RAG variants by 6-10 percentage points. This gap reflects the interpretability-performance tradeoff identified in Chapter 2: machine learning methods can achieve moderate detection accuracy, but they cannot generate the natural language explanations that RAG systems produce. The results demonstrate that this tradeoff is not fundamental—RAG systems achieve both higher accuracy and full interpretability.

7.3 RQ2: Most Competitive Baseline

RQ2 asked which baseline approach shows the greatest competitiveness compared to the proposed framework. The results unambiguously identify non-agentic RAG as the most competitive baseline. On HDFS, non-agentic RAG achieves $F1=0.864$ compared to the agentic system's 0.907—a gap of 4.3 percentage points. Isolation Forest trails by 10.1 points and the rule-based system by 19.4 points. The pattern is identical on BGL. Non-

agentic RAG is not merely closer to the proposed system than other baselines; it is the only baseline that approaches competitive performance.

This finding has theoretical and practical implications. Theoretically, it suggests that the combination of retrieval and generation is more important than the number of reasoning steps. The gap between rule-based and non-agentic RAG (15+ percentage points) is larger than the gap between non-agentic and agentic RAG (4 points). This indicates that retrieval-augmented generation addresses the most consequential limitations of traditional approaches—rigidity and lack of contextual knowledge—and that agentic reasoning provides incremental improvement rather than a qualitative leap. Practically, this suggests that organizations currently using rule-based or machine learning systems could achieve substantial performance gains by adopting even single-pass RAG, and that the additional complexity of agentic reasoning is justified only when the highest possible accuracy is required.

The performance variation across baselines also illuminates what aspects of log analysis are most difficult. Isolation Forest achieves recall values (0.82 on HDFS, 0.79 on BGL) that are competitive with the agentic system's recall (0.89 and 0.86), but its precision is substantially lower (0.79 and 0.76 vs. 0.92 and 0.89). This indicates that statistical feature-based methods can identify most anomalies but generate excessive false positives because they lack the semantic understanding that distinguishes genuine faults from statistical outliers. RAG systems, by retrieving domain knowledge that contextualizes log patterns, achieve the precision that pure statistical methods cannot.

7.4 RQ3: Alert Quality and Actionability

RQ3 asked whether the agentic RAG system demonstrates greater alert specificity compared to the rule-based approach. The qualitative evaluation results provide strong affirmative evidence. Table 7.2 summarizes the dimensional score differences.

Dimension	Agentic RAG	Rule-Based	Difference
Component Identification	4.6/5	3.1/5	+1.5
Root Cause Explanation	4.2/5	1.4/5	+2.8

Recommendation	4.1/5	0.8/5	+3.3
Specificity			

Table 7.2: Alert quality dimensional score comparison

The agentic RAG system achieves an average total score of 12.9/15, placing 85% of alerts in the high actionability category. The rule-based system averages 5.3/15, with 60% of alerts scoring low. The largest gap appears in recommendation specificity (+3.3 points) and root cause explanation (+2.8 points)—the dimensions that require reasoning beyond simple pattern matching. Component identification shows a smaller gap (+1.5 points) because rule-based systems can often extract component names from log templates, but they cannot explain why the component is involved or what should be done about it.

These results validate the argument made by Chen et al. (2023) that anomaly detection systems must be evaluated on the quality of their explanatory outputs, not merely their detection accuracy. A system that achieves high F1-scores but generates uninformative alerts provides limited operational value. The agentic RAG system's ability to produce alerts with detailed reasoning chains, specific component identification, and concrete recommendations directly addresses this gap. The reasoning chains make the system's analytical process auditable, enabling operators to verify the basis of alerts rather than accepting them as opaque verdicts.

7.5 Comparison with Published Literature

Comparing the results obtained here with published figures from the log analysis literature requires careful interpretation, as evaluation methodologies, datasets, and baseline implementations vary across studies. Guo et al. (2021) report F1-scores of 0.982 on HDFS for LogBERT, substantially higher than the 0.907 achieved by the proposed system. However, their evaluation uses a different train/test split and excludes certain log types, making direct comparison problematic. Le and Zhang (2022) demonstrate that LogBERT's performance degrades substantially when evaluated on logs with format drift and label noise—conditions representative of live production environments—achieving F1-scores in the 0.85-0.90 range on realistic test sets. The proposed system's performance of 0.907 on HDFS and 0.871 on BGL falls within this range, suggesting that its

performance is competitive with state-of-the-art deep learning methods under realistic conditions.

More importantly, the proposed system addresses a limitation that LogBERT and other deep learning methods do not: interpretability. LogBERT produces numerical anomaly scores with no accompanying explanation. The proposed system produces reasoning chains and structured alerts that enable operators to understand and act on detected anomalies. This represents a qualitative advance that cannot be captured in F1-score comparisons alone. Chen et al. (2023) argue that the log analysis field has prioritized detection accuracy at the expense of decision support quality, and that future research must treat interpretability as a primary requirement rather than an optional supplement to detection performance. The present research demonstrates that high detection accuracy and full interpretability are not mutually exclusive.

The agentic architecture draws on recent work by Yao et al. (2023) on ReAct and Wang et al. (2023) on LLM-based autonomous agents, but applies these frameworks to a domain—log analysis—where they have not previously been evaluated. Yao et al. (2023) report performance improvements of up to 34% on multi-hop question-answering tasks when using ReAct compared to standard prompting. The 4+ percentage point improvement observed here is more modest, which may reflect that log analysis is a more constrained task than open-domain question answering. Each log entry provides a specific analytical target, and the knowledge base is domain-focused rather than encyclopedic. Nevertheless, the consistent improvement across both datasets provides evidence that the ReAct pattern generalizes beyond the tasks for which it was originally developed.

7.6 Practical Implications

The results have implications for organizations operating distributed systems at scale. First, the substantial false positive rate reduction—from 32.4% for rule-based systems to 7.8% for agentic RAG on HDFS—directly addresses the alert fatigue problem that degrades operator responsiveness in production monitoring environments. A false positive rate below 10% means that more than nine out of ten alerts are genuine, which maintains operator confidence in the monitoring system. Second, the alert quality results

demonstrate that the system produces outputs suitable for immediate operational use. Alerts with average scores of 12.9/15 provide specific component identification, detailed root cause explanations, and concrete recommended actions—the information operators need to respond effectively without conducting additional investigation.

Third, the comparison between agentic and non-agentic RAG provides guidance on when the additional complexity of multi-step reasoning is justified. For organizations prioritizing simplicity and minimal inference latency, single-pass RAG may provide the best cost-benefit tradeoff, achieving F1-scores around 0.85 with simpler implementation. For organizations where detection accuracy is paramount—financial trading systems, healthcare platforms, or critical infrastructure—the 4 percentage point improvement from agentic reasoning may justify the additional computational cost and implementation complexity. The cross-dataset consistency of results provides confidence that these performance characteristics generalize beyond the specific systems evaluated here.

Fourth, the use of open-source components—Mistral 7B served locally through Ollama, FAISS for vector search, and Drain3 for log parsing—demonstrates that effective agentic RAG systems can be built without dependence on proprietary commercial APIs. This reduces operational cost, eliminates external service dependencies, and ensures that sensitive operational logs never leave the organization's infrastructure. The complete system implementation, excluding the LLM weights themselves, consists of under 1,200 lines of Python code, suggesting that adoption barriers are primarily conceptual rather than technical.

7.7 Limitations of This Research

Several limitations must be acknowledged. First, the evaluation is conducted on two benchmark datasets under controlled experimental conditions rather than in live production environments. While both datasets are derived from real systems, they represent snapshots of historical data rather than streaming logs with the temporal dynamics and operational pressures of production monitoring. Second, the LLM component uses Mistral 7B—a model considerably smaller than frontier models available at the time of writing. Larger models might achieve higher performance, but evaluating this claim would require access to substantially greater computational resources.

Third, the qualitative alert quality evaluation for RQ3 was conducted by a single evaluator, introducing the risk of rater bias. Inter-rater reliability assessment with multiple independent evaluators would strengthen confidence in the rubric scores. Fourth, the knowledge base contains 150 documents for HDFS and 120 for BGL, constructed specifically for this research. Real-world deployment would require substantially larger knowledge bases continuously updated as new failure patterns emerge and system components evolve. Fifth, the study does not evaluate inference latency or computational cost, both of which are relevant for production deployment. The multi-step agentic reasoning loop requires multiple LLM calls per log entry, which increases latency compared to single-pass methods.

Chapter 8

CONCLUSION

8.1 Introduction

This chapter concludes the dissertation by summarizing the research contributions, restating the key findings that address each research question, acknowledging the limitations of the work, and proposing directions for future research. The chapter begins by articulating the specific contributions this research makes to the log analysis field before providing a concise summary of findings organized by research question. The chapter concludes by identifying promising avenues for extending and improving the proposed approach.

8.2 Research Contributions

This research makes four distinct contributions to the automated log analysis literature. Table 8.1 summarizes these contributions and their significance.

Contribution	Significance
Novel agentic RAG architecture for log analysis	First application of ReAct-based multi-step reasoning to anomaly detection in system logs

Empirical validation across benchmarks	Systematic comparison on HDFS and BGL with statistical significance testing
Demonstrated interpretability improvement	Quantitative evidence that agentic systems produce more actionable alerts (12.9/15 vs 5.3/15)
Open-source implementation	Complete working prototype using Mistral 7B, FAISS, and Drain3 (1,200 lines of code)

Table 8.1: Summary of research contributions

The first contribution is architectural: the design and implementation of an agentic RAG framework that combines retrieval-augmented generation with multi-step ReAct reasoning for log anomaly detection. While RAG systems have been applied to question answering and code generation, and agentic systems have been evaluated on web navigation and tool use, this research is the first to demonstrate that agentic RAG can improve both detection accuracy and alert quality in system log analysis. The second contribution is empirical: systematic evaluation on established benchmarks with ground-truth labels, statistical significance testing, and comparison with three baselines representing different categories of existing approaches. This provides quantitative evidence that the proposed approach outperforms rule-based, machine learning, and single-pass RAG systems.

The third contribution is methodological: the development and application of a structured rubric for evaluating alert quality across three dimensions—component identification, root cause explanation, and recommendation specificity. This addresses the gap identified by Chen et al. (2023), who argue that detection metrics alone are insufficient for evaluating systems intended to support human decision-making. The fourth contribution is practical: a complete open-source implementation using accessible tools that demonstrates agentic RAG systems can be built without dependence on proprietary commercial APIs.

8.3 Summary of Key Findings

The research produced clear answers to all three research questions. Table 8.2 summarizes the key findings.

Research Question	Finding	Evidence
RQ1: Detection improvement?	Yes - significant improvement	F1: 0.907 (HDFS), 0.871 (BGL); FPR: 7.8% (HDFS); $p < 0.05$ all comparisons
RQ2: Most competitive baseline?	Non-agentic RAG	F1 gap: 4.3pp (HDFS), 4.1pp (BGL); Isolation Forest: 10pp gap; Rule-based: 19pp gap
RQ3: Greater alert specificity?	Yes - substantial improvement	Alert quality: 12.9/15 vs 5.3/15; 85% high-actionability alerts vs 0%

Table 8.2: Summary of findings by research question

RQ1 asked whether the agentic RAG approach improves accuracy for anomaly detection. The answer is affirmative with strong statistical support. The proposed system achieves F1-scores of 0.907 on HDFS and 0.871 on BGL, outperforming all three baselines with p-values below 0.05 in all pairwise comparisons. The false positive rate of 7.8% on HDFS represents a 76% reduction compared to the rule-based baseline, directly addressing the alert fatigue problem that undermines operator responsiveness in production environments.

RQ2 asked which baseline approach shows the greatest competitiveness. Non-agentic RAG emerges as the most competitive baseline, trailing the agentic system by 4+ percentage points on both datasets. This finding has theoretical implications: it demonstrates that retrieval-augmented generation addresses the most consequential limitations of traditional approaches, and that agentic reasoning provides incremental rather than transformational improvement. Organizations seeking to improve log analysis capabilities should prioritize adopting RAG over moving from rule-based to machine learning systems.

RQ3 asked whether the agentic RAG system demonstrates greater alert specificity. The qualitative evaluation provides strong affirmative evidence. The agentic system achieves average scores of 12.9/15, with 85% of alerts rated as highly actionable. The rule-based

system averages 5.3/15, with 60% of alerts rated low. The largest gaps appear in root cause explanation (+2.8 points) and recommendation specificity (+3.3 points)—the dimensions requiring reasoning beyond pattern matching. These results validate the argument that anomaly detection systems must be evaluated on alert quality, not merely detection accuracy.

8.4 Theoretical Implications

The findings contribute to theoretical understanding of how retrieval-augmented generation and agentic reasoning improve performance on knowledge-intensive tasks. The consistent 4+ percentage point improvement of agentic over non-agentic RAG across structurally different datasets provides empirical support for the claim made by Yao et al. (2023) that interleaved reasoning and acting outperforms single-step generation. The theoretical mechanism is that multi-step reasoning enables the system to refine its understanding iteratively—retrieving knowledge, assessing whether additional context is needed, and adjusting its analysis based on what it learns. This aligns with human analytical processes more closely than single-pass inference.

The finding that non-agentic RAG substantially outperforms machine learning baselines (6-10 percentage points) suggests that semantic understanding—the ability to reason about the meaning of log content rather than merely its statistical properties—is more valuable for anomaly detection than sophisticated statistical modeling. This has implications beyond log analysis for any domain where context and domain knowledge are required to interpret observations correctly. Tasks characterized by high semantic complexity and availability of relevant external knowledge are prime candidates for RAG-based approaches.

8.5 Practical Implications

For practitioners operating distributed systems, the findings suggest that agentic RAG systems can deliver detection performance competitive with state-of-the-art deep learning methods while providing the interpretability that machine learning approaches lack. The 7.8% false positive rate achieved on HDFS is low enough to maintain operator trust in the monitoring system, while the detailed reasoning chains enable operators to verify alert validity before taking action. The use of open-source components—Mistral 7B, FAISS,

Drain3—demonstrates that effective systems can be built without commercial API dependencies, reducing operational cost and ensuring sensitive logs never leave organizational infrastructure.

The comparison between agentic and non-agentic RAG provides decision guidance. Organizations prioritizing simplicity may find single-pass RAG offers the best cost-benefit tradeoff, achieving F1-scores around 0.85 with minimal implementation complexity. Organizations where detection accuracy is paramount—financial systems, healthcare platforms, critical infrastructure—may find the 4 percentage point improvement from agentic reasoning justifies the additional computational cost. The complete implementation in under 1,200 lines of Python code suggests adoption barriers are primarily conceptual rather than technical.

8.6 Limitations and Future Work

The research has several limitations that suggest directions for future work. Table 8.3 summarizes the limitations, and Table 8.4 presents corresponding future research directions.

Limitation	Impact
Evaluation on benchmarks only	Real-world performance may differ due to temporal dynamics and operational pressures
Single LLM (Mistral 7B)	Larger models might achieve higher performance
Single-rater alert evaluation	Rater bias may affect rubric scores
Static knowledge bases	Real deployment requires continuous knowledge base updates
No latency/cost analysis	Multi-step reasoning increases inference time and computational cost

Table 8.3: Research limitations and their impacts

Future Research Direction	Research Questions
Live production deployment	Does performance hold under streaming logs with

	operational pressures?
Comparison with frontier LLMs	How much does performance improve with GPT-4, Claude 3, or Llama 3 70B?
Multi-rater alert evaluation	What is inter-rater reliability on the alert quality rubric?
Adaptive knowledge base updates	Can the system automatically update its knowledge base from incident reports?
Latency optimization	Can caching, batching, or speculative execution reduce inference time?

Table 8.4: Future research directions

The most pressing direction for future work is live production deployment and evaluation under realistic operational conditions. Benchmark datasets represent snapshots of historical data, but production systems generate streaming logs with temporal dependencies, evolving failure patterns, and operational pressures that static evaluations cannot capture. A six-month production trial comparing the agentic RAG system against the existing monitoring infrastructure would provide stronger evidence of real-world effectiveness. Second, systematic comparison across multiple LLM backends—including frontier models like GPT-4, Claude 3, and Llama 3 70B—would determine how much performance depends on model scale versus architectural design. Third, expanding the alert quality evaluation to multiple independent raters would strengthen confidence in the rubric scores and enable inter-rater reliability assessment.

8.7 Concluding Remarks

This dissertation has demonstrated that agentic retrieval-augmented generation can improve both detection accuracy and alert actionability in system log analysis. The proposed framework achieves F1-scores of 0.907 on HDFS and 0.871 on BGL, outperforming rule-based, machine learning, and single-pass RAG baselines with statistical significance. Alert quality scores of 12.9/15 demonstrate that the system produces outputs suitable for operational use. The findings contribute to theoretical understanding of how multi-step reasoning improves performance on knowledge-

intensive tasks and provide practical guidance for organizations seeking to improve their log monitoring capabilities. The open-source implementation demonstrates that effective agentic systems can be built with accessible tools. While several limitations remain—particularly the need for live production validation—the results provide strong evidence that agentic RAG represents a promising direction for automated log analysis research and practice.

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APPENDICES

Appendix A: Sample Test Cases and System Outputs

This appendix presents representative test cases from the HDFS and BGL datasets, showing raw log entries, preprocessed output, system classifications, and generated alerts. These examples demonstrate the practical operation of the agentic RAG framework and illustrate the differences in output quality between the proposed system and baseline approaches.

A.1 Test Case 1: HDFS Block Replication Failure

Raw Log Entry:

```
081109 203615 148 INFO dfs.DataNode$PacketResponder: PacketResponder 1
for block blk_-1608999687919862906 terminating

081109 203615 148 ERROR dfs.DataNode$DataXceiver: DataXceiver:
java.io.IOException: Block blk_-1608999687919862906 does not exist
```

```
081109 203616 149 WARN dfs.DataNode$BlockReceiver: PacketResponder:
blk_-1608999687919862906, type=TRANSFER_BLOCK, Exception:
java.net.SocketException: Connection reset
```

Preprocessed Output:

Template ID	T_0547
Severity	ERROR
Component	DataNode
Template	DataXceiver: java.io.IOException: Block <*> does not exist
Variables	blk_-1608999687919862906
Timestamp	2008-11-09T20:36:15

Agentic RAG Classification & Alert:

Classification	ANOMALOUS
Confidence	0.94
Reasoning Steps	3 iterations
Component	HDFS DataNode (Block Replication Service)
Root Cause	Block replication failure due to DataNode unavailability. The target DataNode terminated the PacketResponder before block transfer completed, causing IOException. Connection reset indicates network-level failure during block transfer.
Severity	CRITICAL
Impact	Data under-replication. If multiple replicas fail, data loss may occur.
Recommendation	1. Check DataNode health status (dfs.datanode.address) 2. Verify network connectivity between

	DataNodes 3. Review DataNode logs for failure cause 4. Initiate manual block re-replication if needed
Retrieved Knowledge	HDFS replication protocol documentation, DataNode failure patterns, Block transfer error handling procedures

Rule-Based System Output (Comparison):

Classification	ANOMALOUS
Matched Keywords	ERROR, IOException, Exception
Alert	ERROR detected in log entry: DataXceiver IOException
Root Cause/Recommendation	None provided

Ground Truth: ANOMALOUS

Alert Quality Score (Agentic RAG): 14/15 (Component: 5/5, Root Cause: 5/5,
Recommendation: 4/5)

A.2 Test Case 2: BGL Hardware Fault

Raw Log Entry:

2005-06-03-15.42.50.675872 R02-M1-N0-C:J12-U11 2005-06-03-
15.42.50.675872 R02-M1-N0-C:J12-U11 RAS KERNEL FATAL ciod: Error
reading message prefix on CioStream socket to 172.16.96.6:1

Preprocessed Output:

Template ID	T_1243
Severity	FATAL
Component	RAS KERNEL
Template	ciod: Error reading message prefix on CioStream socket to <*>
Variables	172.16.96.6:1
Timestamp	2005-06-03T15:42:50.675872

Agentic RAG Classification & Alert:

Classification	ANOMALOUS
Confidence	0.97
Reasoning Steps	4 iterations
Component	RAS Kernel - Communication I/O Daemon (Node R02-M1-N0-C:J12-U11)
Root Cause	Communication failure between compute node and I/O node. The ciod (compute I/O daemon) cannot read message headers from the I/O node at 172.16.96.6, indicating either network failure, I/O node crash, or socket corruption. FATAL severity indicates unrecoverable error.
Severity	FATAL

Impact	Node R02-M1-N0-C:J12-U11 unable to perform I/O operations. Running jobs on this node will fail.
Recommendation	1. Check I/O node 172.16.96.6 status 2. Verify network connectivity to compute node 3. Review RAS logs for I/O node failures 4. Drain compute node and mark for maintenance 5. Restart cioid service if I/O node is healthy
Retrieved Knowledge	BGL RAS event categories, cioid error patterns, I/O node communication protocol

Ground Truth: ANOMALOUS

Alert Quality Score (Agentic RAG): 13/15 (Component: 5/5, Root Cause: 4/5, Recommendation: 4/5)

A.3 Test Case 3: Normal HDFS Operation

Raw Log Entry:

```
081109 203518 143 INFO dfs.DataNode$DataXceiver: Receiving block blk_-
1608999687919862906 src: /10.251.73.220:54106 dest:
/10.251.73.220:50010
```

```
081109 203519 143 INFO dfs.DataNode$PacketResponder: PacketResponder 0
for block blk_-1608999687919862906 terminating
```

Preprocessed Output:

Template ID	T_0012
Severity	INFO
Component	DataNode
Template	Receiving block <*> src: <*> dest: <*>
Variables	blk_-1608999687919862906, /10.251.73.220:54106, /10.251.73.220:50010

Agentic RAG Classification & Alert:

Classification	NORMAL
Confidence	0.91
Reasoning Steps	2 iterations
Analysis	Standard block replication operation. Block received successfully and PacketResponder terminated normally. INFO severity, successful completion pattern, normal termination sequence all indicate healthy operation.
Action	No alert generated

Ground Truth: NORMAL

Appendix B: Sample Knowledge Base Documents

This appendix presents representative documents from the knowledge base used by the agentic RAG framework. These documents are organized into four categories: error patterns, component documentation, incident reports, and remediation actions.

B.1 Error Pattern Document: HDFS Block Replication Failure

Document Category: Error Patterns

Document ID: EP_HDFS_001

Content:

HDFS Block Replication Failure Pattern

Description: This failure pattern occurs when a DataNode is unable to complete block replication to another DataNode. The typical signature includes:

- IOException with message 'Block does not exist'
- PacketResponder termination before completion
- SocketException with 'Connection reset'

Common Causes:

1. Target DataNode crashed or became unavailable during transfer
2. Network connectivity issues between DataNodes
3. Disk failure on target DataNode
4. DataNode process killed during block transfer

Impact: Under-replication of data blocks. If replication factor drops below configured minimum, data loss risk increases. Critical for blocks with replication factor of 1.

Related Components: DataNode, PacketResponder, DataXceiver, Block Transfer Protocol

B.2 Component Documentation: BGL RAS Kernel

Document Category: Component Documentation

Document ID: CD_BGL_002

Content:

BlueGene/L RAS Kernel Component

Description: The Reliability, Availability, and Serviceability (RAS) kernel is the central component responsible for monitoring system health and reporting hardware and software events on BlueGene/L supercomputers.

Key Responsibilities:

- Collecting events from compute nodes, I/O nodes, and system components
- Classifying events by severity (INFO, WARNING, SEVERE, FATAL)
- Managing communication between compute nodes and I/O nodes via ciod
- Logging hardware failures, network errors, and system anomalies

ciod (Compute I/O Daemon): Mediates I/O operations between compute nodes and I/O nodes. Failures in ciod indicate communication breakdowns that prevent compute nodes from performing file I/O operations.

Event Severity Levels:

- INFO: Normal operations, informational messages
- WARNING: Potential issues that do not impact functionality
- SEVERE: Errors requiring attention, may impact performance
- FATAL: Unrecoverable errors requiring intervention

B.3 Remediation Action: DataNode Block Replication Recovery

Document Category: Remediation Actions

Document ID: RA_HDFS_003

Content:

Remediation Procedure: HDFS Block Under-Replication Recovery

Symptoms: Blocks showing replication factor below configured minimum, DataNode communication failures during block transfer.

Immediate Actions:

1. Identify affected blocks using: `hdfs fsck / -files -blocks -locations`
2. Check DataNode health: `hdfs dfsadmin -report`
3. Verify network connectivity between DataNodes
4. Review DataNode logs for error patterns

Recovery Steps:

1. If DataNode is healthy but network failed: Wait for network recovery, HDFS will automatically re-replicate
2. If DataNode crashed: Restart DataNode service, monitor block replication progress
3. If DataNode disk failed: Decommission failed DataNode, add replacement, trigger rebalancing
4. Manual replication trigger (if needed): `hdfs dfsadmin -triggerBlockReport <datanode>`

Verification:

- Run fsck again after 10 minutes to verify replication recovery
- Monitor NameNode logs for under-replicated block count
- Check that replication factor returns to configured level

Appendix C: Alert Quality Rubric Evaluation Examples

This appendix presents detailed examples of alert quality evaluation using the rubric from Table 3.4. Two alerts are shown: one from the agentic RAG system and one from the rule-based baseline, both responding to the same anomalous log entry.

C.1 Example Alert Evaluation: HDFS Block Replication Failure

Log Entry: DataXceiver: java.io.IOException: Block blk_-1608999687919862906 does not exist

Ground Truth: ANOMALOUS (DataNode failure during block replication)

Agentic RAG System Alert:

— ALERT —

Timestamp: 2024-04-15 14:32:17

Severity: CRITICAL

Component: HDFS DataNode - Block Replication Service

Classification: ANOMALOUS

Confidence: 0.94

Root Cause:

Block replication failure due to DataNode unavailability. The target DataNode terminated the PacketResponder before block transfer completed, causing IOException. Connection reset indicates network-level failure during block transfer.

Recommended Actions:

1. Check DataNode health status using: `hdfs dfsadmin -report`
2. Verify network connectivity between source and target DataNodes
3. Review DataNode logs at target node for failure cause
4. If DataNode is down, restart service and monitor replication recovery
5. Initiate manual block re-replication if needed: `hdfs dfsadmin -`

triggerBlockReport

Impact: Block blk_-1608999687919862906 is under-replicated. Data loss risk if additional replicas fail.

Rubric Evaluation (Agentic RAG):

Dimension	Score	Justification
Component Identification	5/5	Specific component named: "HDFS DataNode - Block Replication Service". Contextualized with block ID.
Root Cause Explanation	5/5	Detailed causal chain: DataNode termination → PacketResponder failure → IOException. Network-level failure identified.
Recommendation Specificity	4/5	Concrete commands provided (hdfs dfsadmin). Step-by-step procedure. Minor: Could specify exact log file paths.
TOTAL	14/15	High actionability. Operator can act immediately without further investigation.

Rule-Based System Alert:

— ALERT —

Severity: ERROR
Matched Keyword: IOException
Log Template: DataXceiver: java.io.IOException: Block <*> does not exist
Classification: ANOMALOUS

Rubric Evaluation (Rule-Based):

Dimension	Score	Justification
Component Identification	3/5	Component mentioned ("DataXceiver") but not contextualized. No explanation of role or subsystem.
Root Cause Explanation	1/5	No root cause provided. Only reports matched keyword "IOException". No causal reasoning.
Recommendation Specificity	0/5	No recommendations provided. Operator must investigate independently.
TOTAL	4/15	Low actionability. Alert provides detection but no guidance for response.

Comparative Summary:

The agentic RAG alert achieves a score of 14/15 (93%) compared to the rule-based alert's 4/15 (27%). The key differences are:

1. Component Context: Agentic RAG identifies the specific service ('Block Replication Service') while rule-based only reports the class name.
2. Root Cause: Agentic RAG provides a complete causal explanation with evidence from the log sequence. Rule-based provides none.
3. Actionability: Agentic RAG provides concrete commands and a step-by-step procedure. Rule-based requires the operator to determine appropriate actions independently.

This 10-point gap (14 vs 4) is representative of the average difference observed across the 20-alert evaluation sample (12.9 vs 5.3).